

# Processes and flows in Power Apps and Power Automate

## 1. Introduction to workflows and processes

Workflows and processes in Power Apps and Power Automate are important for several reasons, contributing to the efficiency, consistency, and productivity of organizations.

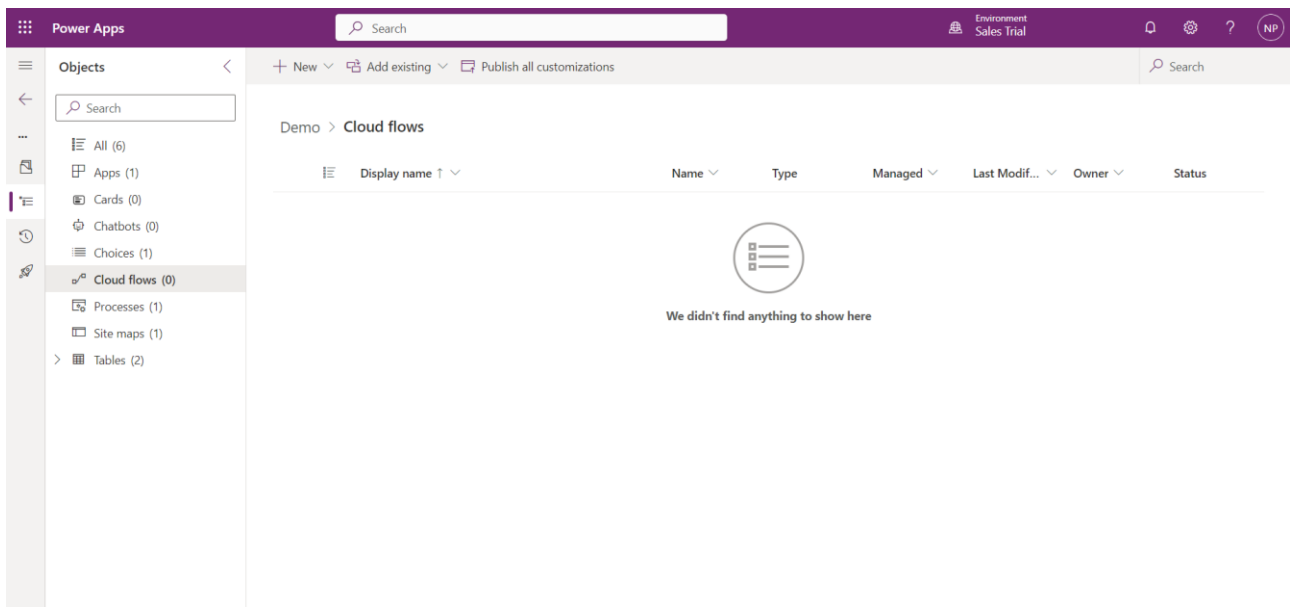
Here are some key reasons why they are considered crucial:

1. **Automation of Repetitive Tasks:** workflows and processes enable the automation of routine and repetitive tasks. This reduces manual effort, minimizes errors, and allows employees to focus on more strategic and value-added activities.
2. **Efficiency Improvement:** by automating business processes, workflows and processes streamline operations, making them more efficient. Tasks are executed automatically, reducing the time and resources required to complete them manually.
3. **Consistency and Standardization:** workflows and processes help enforce consistency and standardization in business processes. This ensures that tasks are performed in a uniform manner, reducing variability and improving the overall quality of work.
4. **Enhanced Collaboration:** automation tools facilitate better collaboration among team members. Workflows can be designed to trigger actions based on the completion of tasks by others, promoting seamless handovers and collaboration across different departments.
5. **Reduced Human Error:** automation reduces the risk of human error that can occur in manual processes. Automated workflows and processes follow predefined rules and logic, minimizing the chances of mistakes or oversights.
6. **Faster Decision-Making:** with automated processes, information flows more quickly through the organization. This allows for faster decision-making as relevant data is made available in a timely manner, enabling teams to respond promptly to changing circumstances.
7. **Improved Compliance and Governance:** workflows and processes can be configured to enforce compliance with regulatory requirements and organizational policies. This ensures that business processes adhere to industry standards and legal guidelines.
8. **Empowerment of Citizen Developers:** the low-code/no-code nature of Power Apps and Power Automate empowers non-developers (citizen developers) to create and modify workflows and processes. This democratizes the ability to implement automation solutions without relying heavily on IT departments.
9. **Adaptability to Change:** workflows and processes can be easily modified and adapted to changing business needs. This agility allows organizations to respond quickly to new requirements, market conditions, or internal changes.
10. **Integration with Other Services:** Power Apps and Power Automate can seamlessly integrate with other Microsoft and third-party services. This integration capability allows organizations to connect various systems and create end-to-end solutions that span multiple platforms.

In summary, workflows and processes play a crucial role in digital transformation by automating and optimizing business processes. They contribute to operational efficiency, consistency, and the overall agility of an organization.

To explore the processes of Power Apps and the flows of Power Automate, we will be connecting to the tables that we have created previously. Before moving to Power Automate, which is the modern environment into which we should create the flows, we first see how we can create some processes in Dynamics 365 directly from Power Apps.

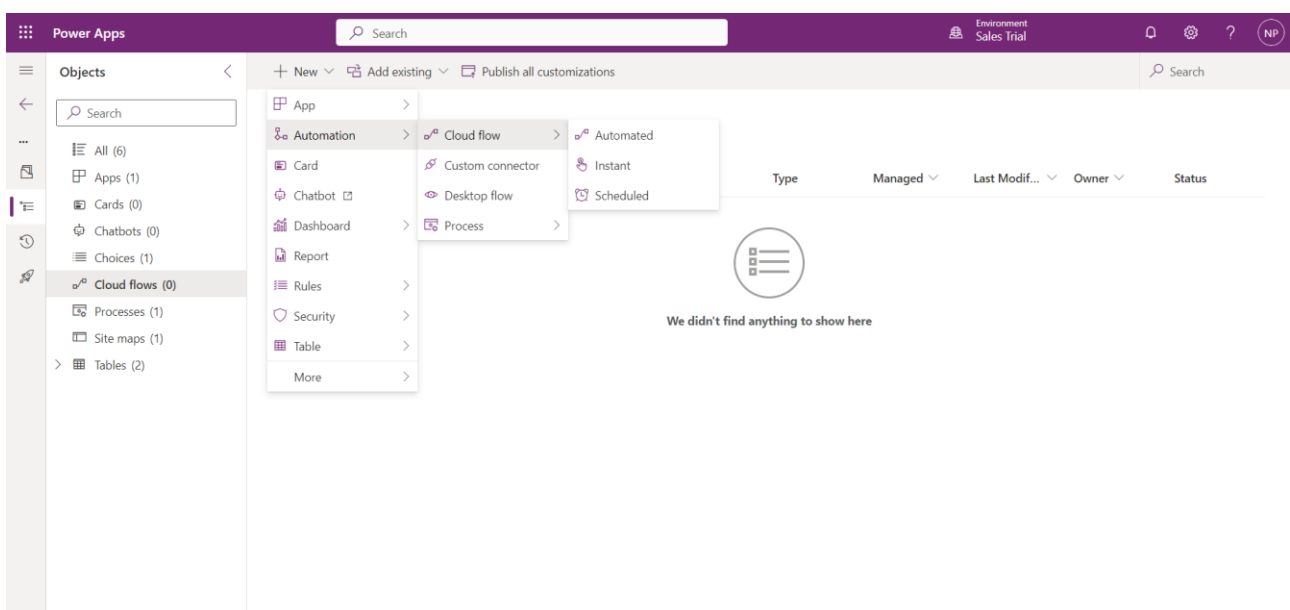
Navigate to Demo > All in Power Apps. On the left panel we see Cloud flows, which are nothing but Power Automate flows (accessible as well from [make.powerautomate.com](https://make.powerautomate.com)).



Power Automate helps us in creating flows and connect to any external API. Every connector has its own authentication and protocol to be followed, but Microsoft gives us a direct connector, namely Dataverse, for connecting to the CRM.

## 2. Create processes in Power Apps

Let us create our first automated cloud flow in Power Apps. Click on Cloud flows and select + New, Automation and Cloud flow.

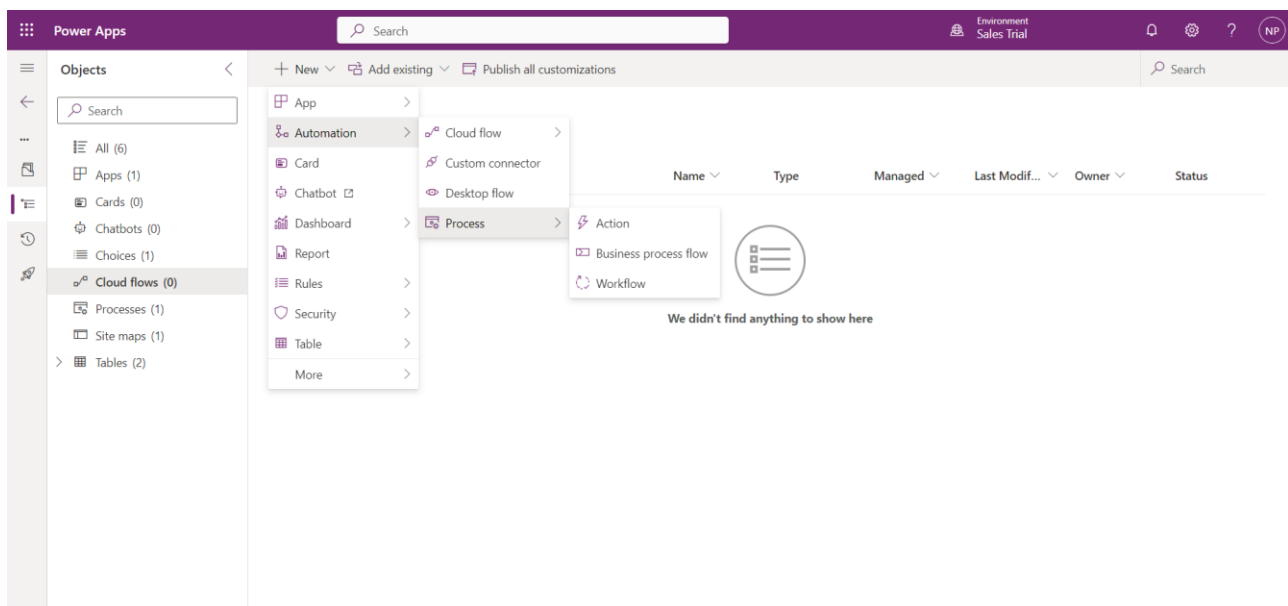


An **automated cloud flow** is triggered by an event that occurs in a connected application or service. When the specified trigger condition is met, the flow is automatically initiated. For instance, an automated cloud flow could be triggered when a new email arrives in a specified mailbox, when a file is added to a cloud storage service, or when a record is created or updated in a database.

A **scheduled cloud flow** is designed to run at specific intervals, such as daily, weekly, or monthly. It is not triggered by a specific event but is set to run on a predefined schedule. For instance, a scheduled cloud flow might be configured to run every day at a certain time to perform a specific task, like sending a daily summary email or updating records in a database.

An **instant cloud flow**, formerly known as "manual" flow, is initiated manually by a user. It allows users to trigger a flow whenever they need to execute a specific task or process. For instance, an instant cloud flow could be triggered by a button click in a mobile app or a web page, allowing users to start a flow on-demand. This type of flow is useful when automation is required but doesn't follow a predictable schedule or event trigger.

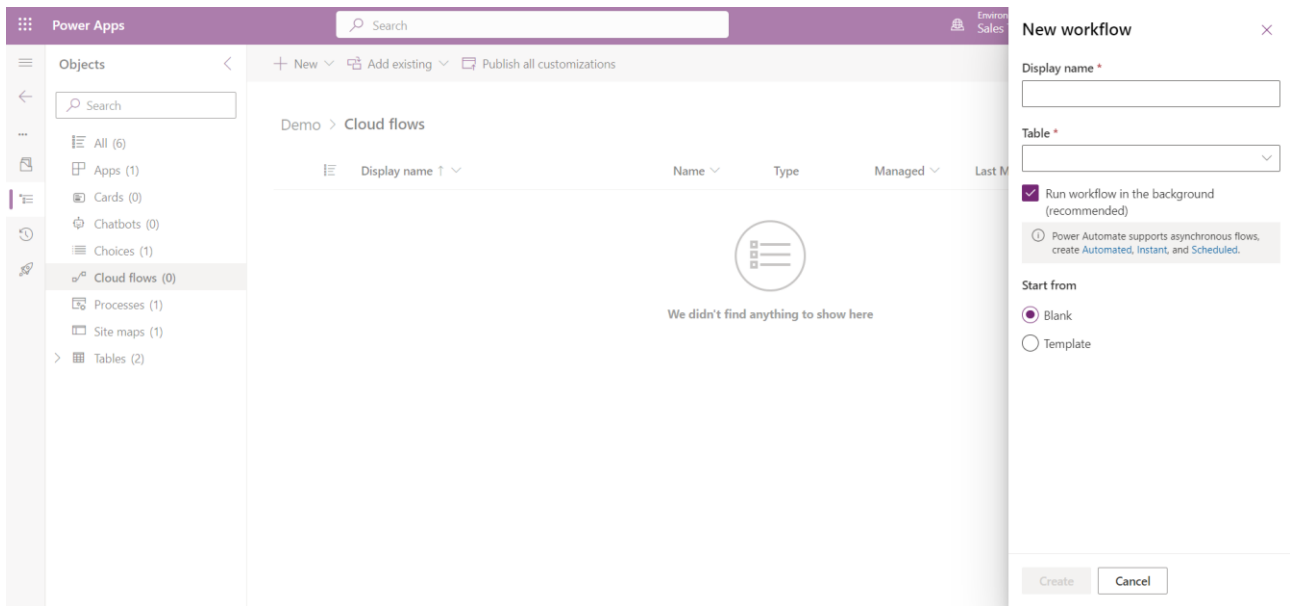
If we hover the mouse over Process, three options are displayed.



**Action:** a type of process with input and output parameters used to perform some kind of operation.

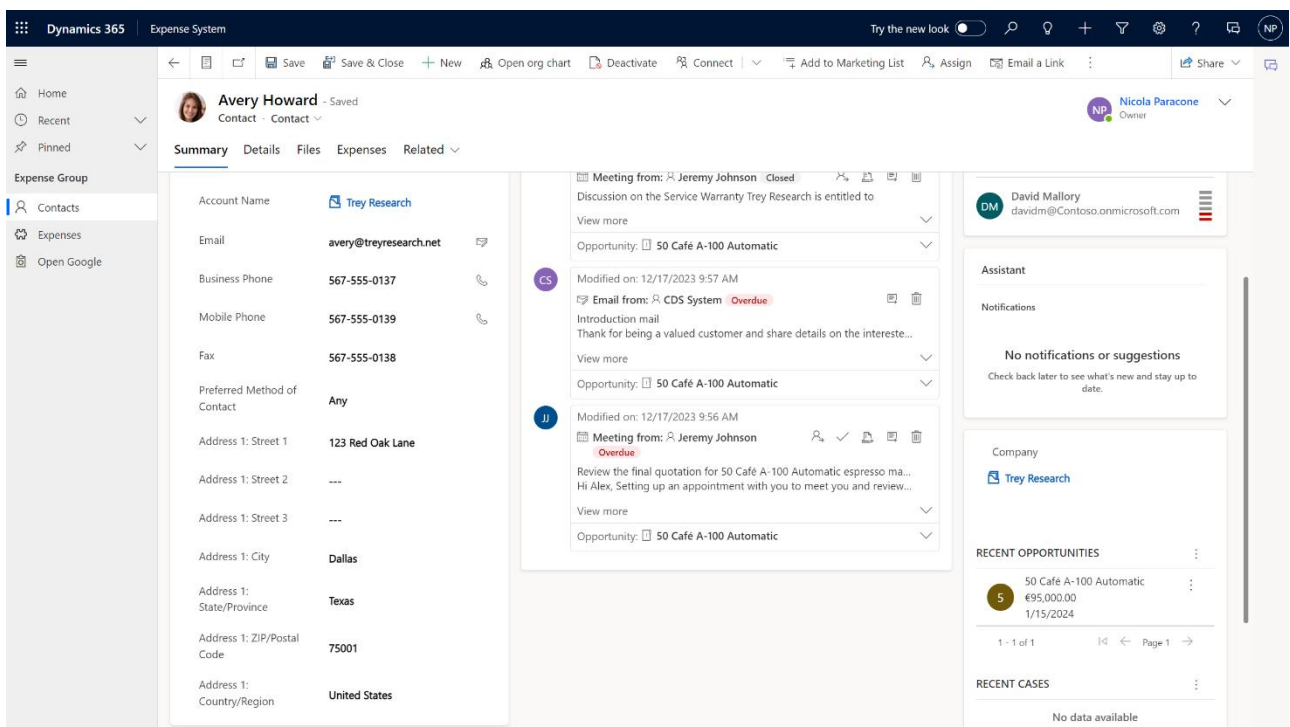
**Business process flow:** a type of processes used to offer a guide for people to get work done. They provide a streamlined user experience that leads people through the processes their organization has defined for interactions that need to be advanced to a conclusion of some kind. This user experience can be tailored so that people with different security roles can have an experience that best suits the work they do. Business process flows define a set of steps for people to follow to take them to a desired outcome. These steps provide a visual indicator that tells people where they are in the business process. Business process flows reduce the need for training because new users don't have to focus on which table they should be using. They can let the process guide them. We can configure business process flows to support common sales methodologies that can help our sales groups achieve better results. For service groups, business process flows can help new staff get up-to-speed more quickly and avoid mistakes that could result in unsatisfied customers.

**Workflow:** yet another type of process designed to accomplish a specific task.



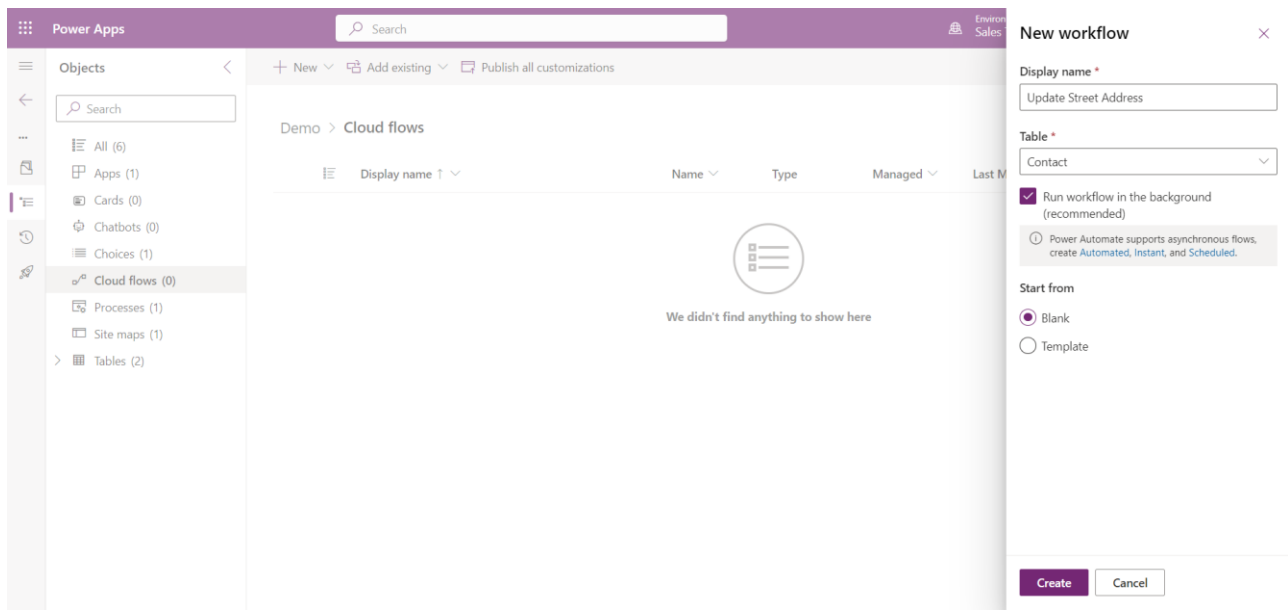
There are two types of workflows: **real-time** and **background** workflows. While background workflows operate without impacting the user's experience, real-time workflows directly affect what the user sees. For instance, consider a scenario where we have two fields, with field 1 containing one value and field 2 containing another. If we need to perform an operation on these fields and display the result to the user immediately, a real-time workflow is necessary. On the other hand, if we use a background flow for the same task, the user will have to refresh the screen multiple times before observing the updated result.

We create a workflow named Update Street Address. If we open any contact, we see that there are fields such as Address 1: Street 1 and Address 1: Street 2.

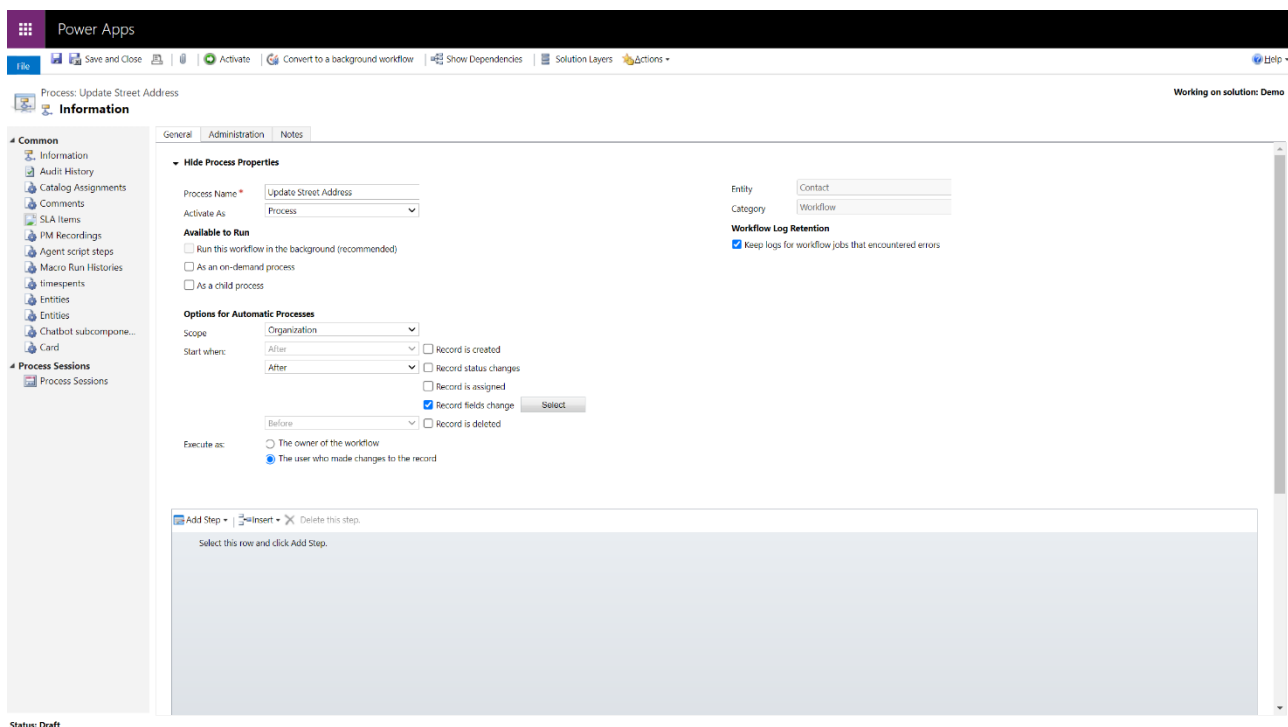


Our aim is to update Address 1: Street 2 with the content of Street 1, City, and State/Province.

So, when creating the workflow, we select Table = Contact.

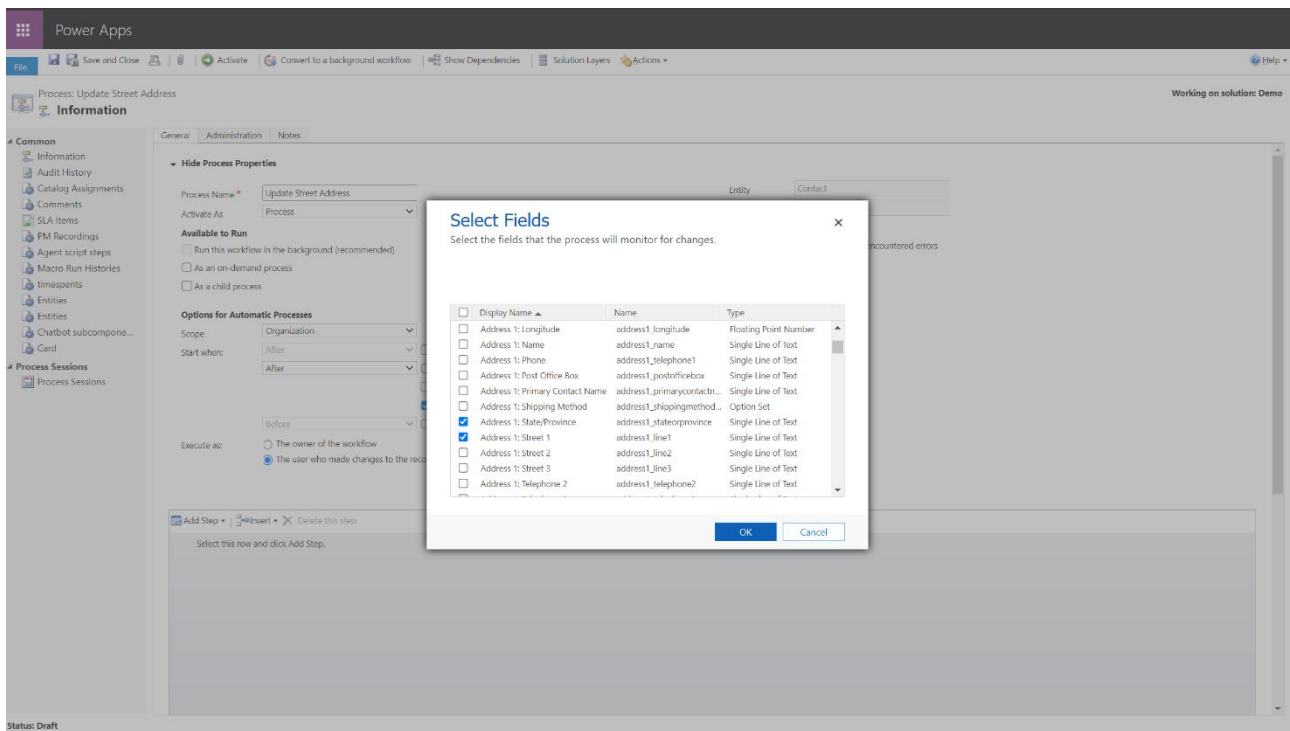


Notice that we can select Run workflow in the background. At present we choose it to be a real-time workflow. Once we click on Create, the workflow designer will pop up. This process will run after it is saved and activated (notice the green button Activate). In the workflow designer we have the possibility to convert a real-time process into a background process and vice versa (the flow must be deactivated to do this operation).



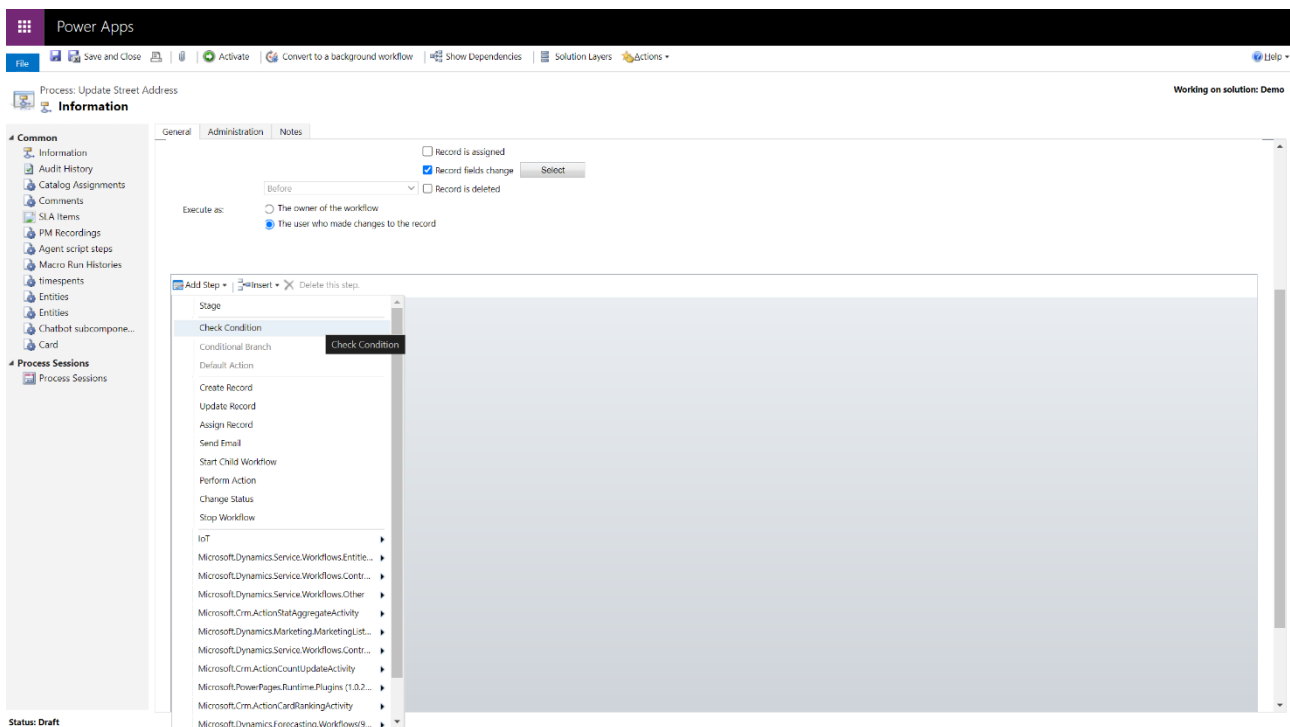
Under Scope we have User, Business Unit, Parent: Child Business Unit and Organization. Here we choose for who we run the process. In our case we choose Organization, as we want the process to run for any user in our organization. Then we have to decide when the process runs. We select Record fields change and then choose the fields that the process will monitor for changes.

Hence, we mark Address 1: Street 1, Address 1: State/Province and Address 1: City.



At this point we define what the workflow will do by adding steps.

Let us assume that a condition has to be checked.



The requirement is the following: populate Street 2 with data from Street 1 + City + State/Province. If any value is null for Street 1 or City or State/Province, throw an error saying "mandatory fields are missing" and stop the workflow. Else update the contact and set the value.

Then we configure the condition. As Primary Entity we choose Contact, which is the entity which we are acting upon. This also gives the related records.

Specify Condition - Microsoft Dynamics 365 - Google Chrome

org8fe6a4bc.crm4.dynamics.com/Condition/Condition.aspx?EntityId=e6b6fd3c-827c-ee11-8179-000d3abbfadf&StepId=ConditionBranchStep2

File Save and Close Help

Clear Group AND Group OR

Contact Select

Select

Form Assistant

Dynamic Values

Dynamic Values

Look for: Contact

Next, we set our requirements grouped under an OR operator.

Specify Condition - Microsoft Dynamics 365 - Google Chrome

org8fe6a4bc.crm4.dynamics.com/Condition/Condition.aspx?EntityId=e6b6fd3c-827c-ee11-8179-000d3abbfadf&StepId=ConditionBranchStep2

File Save and Close Help

Clear Group AND Group OR

OR

|         |                           |                       |
|---------|---------------------------|-----------------------|
| Contact | Address 1: City           | Does Not Contain Data |
| Contact | Address 1: Street 1       | Does Not Contain Data |
| Contact | Address 1: State/Province | Does Not Contain Data |

Select

Form Assistant

Dynamic Values

Dynamic Values

Look for: Contact

(Deprecated) Traversed Path

- Address 1
- Address 1: City
- Address 1: Country/Region
- Address 1: County
- Address 1: Fax
- Address 1: Name
- Address 1: Phone
- Address 1: Post Office Box
- Address 1: Primary Contact Name
- Address 1: State/Province
- Address 1: Street 1
- Address 1: Street 2
- Address 1: Street 3
- Address 1: Telephone 2
- Address 1: Telephone 3
- Address 1: UPS Zone
- Address 1: ZIP/Postal Code
- Address 2
- Address 2: City

Click Save and Close.

So, if this condition is satisfied what should happen? The workflow must stop with a message.

Power Apps

File Save and Close Activate Convert to a real-time workflow Show Dependencies Solution Layers Actions Help

Process: Update Street Address Working on solution: Demo

We recommend using [Microsoft Flow](#) instead of background workflows. [Click here](#) to start building Flows!

General Administration Notes

Add Step Insert Delete this step.

Type a step description here.

If Contact:Address 1: City does not contain data OR Contact:Address 1: Street 1 does not contain data OR Contact:Address 1: State/Province does not contain data, then:

Type a step description here.

Stop workflow with status of: Canceled Set Properties

Status: Draft

Click on Set Properties and insert the message.

Edit Stop Workflow Step Parameters - Microsoft Dynamics 365 - Google Chrome

org8fe6a4bc.crm4.dynamics.com/SFA/Workflow/statusMessageForm.aspx?workflowid=%7be686fd3c-827c-ee11-8179-000d3a8bfadf%7d&activityname=StopWorkflowStep3&readonlymode=false&appSolutionId=%7b42e816a4...

File Save and Close Help

Process: Update Street Address Working on solution: Demo

**Edit Stop Workflow Step Parameters**

| Property Name  | Data Type           | Required | Value                        |
|----------------|---------------------|----------|------------------------------|
| Status message | Single Line of Text | Optional | Mandatory fields are missing |

Form Assistant

Dynamic Values

Dynamic Values

Operator:

Set to

Look for:

Contact

(Deprecated) Traversed Path

Add

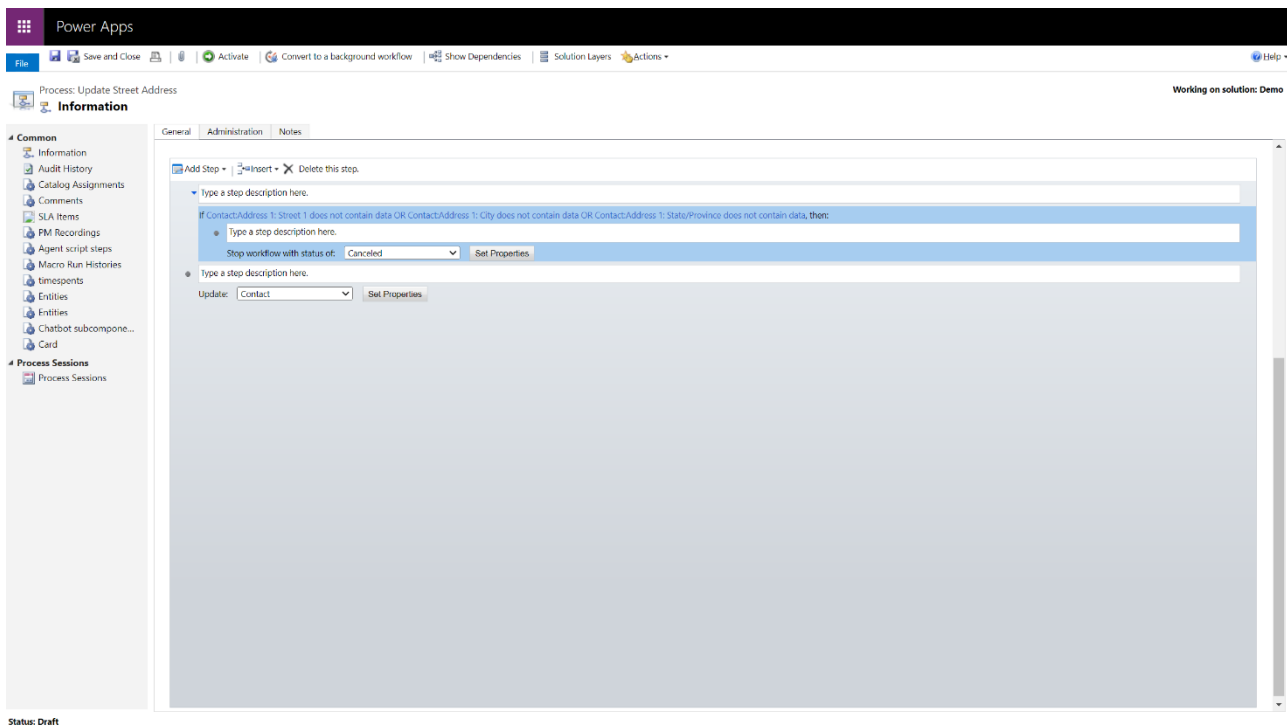
Default value:

OK

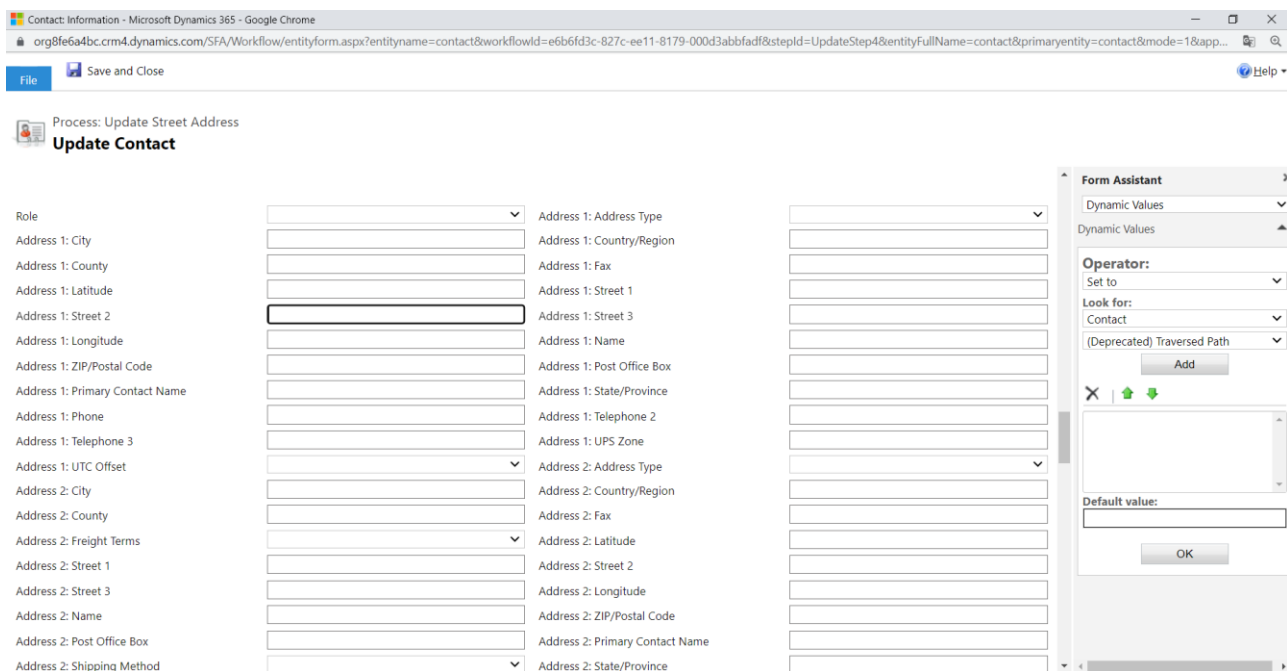
Click on Save and Close.



For the else condition we have to set an Update Record step.



Click on Set Properties and find the field which is to be updated.



On the right we have a menu in which we specify that we look for Contact and obtain all the data related to Contact.

From this panel we populate the Address 1: Street 2 field.

Process: Update Street Address  
**Update Contact**

Additional Fields

Role

Address 1: City

Address 1: Country

Address 1: Latitude

Address 1: Street 2

Address 1: Longitude

Address 1: ZIP/Postal Code

Address 1: Primary Contact Name

Address 1: Phone

Address 1: Telephone 3

Address 1: UTC Offset

Address 2: City

Address 2: Country

Address 2: Freight Terms

Address 2: Street 1

Address 2: Street 3

Address 2: Name

Address 2: Post Office Box

Address 2: Shipping Method

Address 2: Telephone 1

Address 2: Telephone 3

Address 2: UTC Offset

Address 3: City

Address 3: Country

Address 3: Freight Terms

Address 3: Street 1

Address 3: Street 3

Address 3: Name

Address 3: Post Office Box

Address 3: Shipping Method

Address 3: Telephone1

Address 3: Telephone3

Address 3: UTC Offset

Address 1: Address Type

Address 1: Country/Region

Address 1: Fax

Address 1: Street 1

Address 1: Street 3

Address 1: Name

Address 1: Post Office Box

Address 1: State/Province

Address 1: Telephone 2

Address 1: UPS Zone

Address 2: Address Type

Address 2: Country/Region

Address 2: Fax

Address 2: Latitude

Address 2: Street 2

Address 2: Longitude

Address 2: ZIP/Postal Code

Address 2: Primary Contact Name

Address 2: State/Province

Address 2: Telephone 2

Address 2: UPS Zone

Address 3: Address Type

Address 3: Country/Region

Address 3: Fax

Address 3: Latitude

Address 3: Street 2

Address 3: Longitude

Address 3: ZIP/Postal Code

Address 3: Primary Contact Name

Address 3: State/Province

Address 3: Telephone2

Address 3: UPS Zone

Confirm Remove Password

Form Assistant

Dynamic Values

Operator:

Set to

Look for:

Contact

Address 1: State/Province

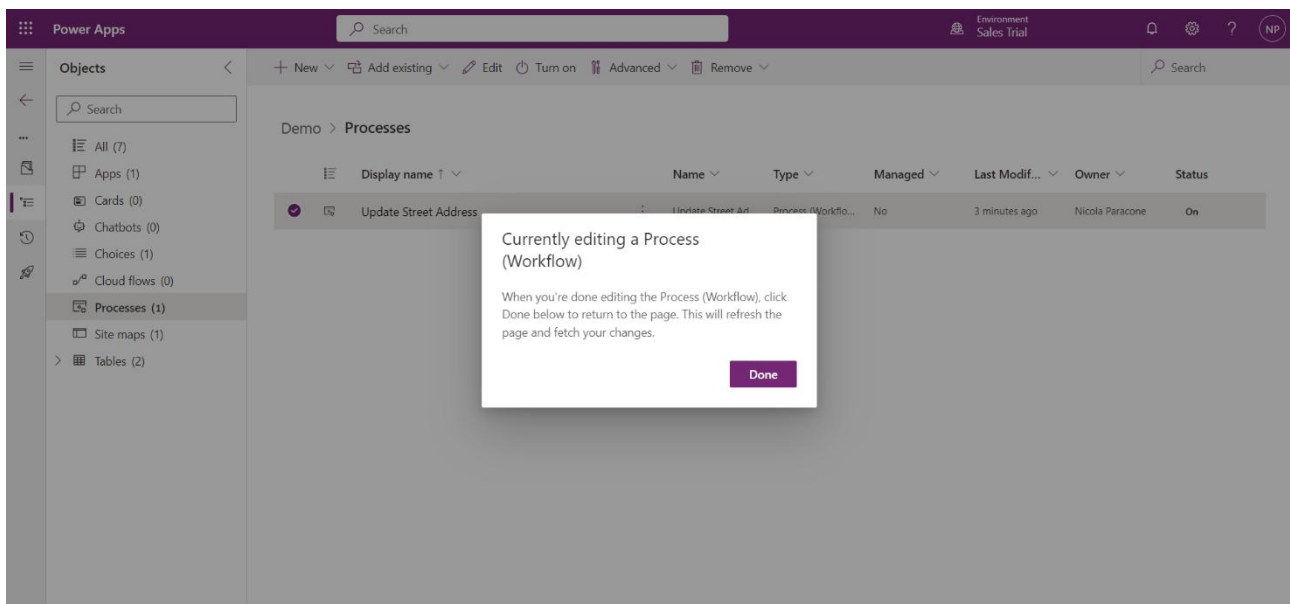
Add

Address 1: State/Province(Contact)

Default values:

OK

Once we are back into the workflow designer, we click on Save and Activate. Since we are done with creating the process, we can click Done in Power Apps.



Now we can test the process.

Open any contact, update the content of Address 1: Street 1 and save.

If the process we created is working correctly, Address 1: Street 2 will be populated with Street 1 + City + State/Province.

The screenshot shows the Microsoft Dynamics 365 Expense System interface. The contact record for Avery Howard is displayed. The 'Address 1: Street 2' field is populated with '123 Red Oak Lane Dallas Texas'. The 'Recent' tab shows a meeting from Jeremy Johnson and an email from CDS System. The 'Assistant' section shows no notifications or suggestions. The 'Company' section shows Trey Research. The 'Recent Opportunities' section shows a 50 Cafe A-100 Automatic opportunity for \$95,000.00, dated 1/15/2024.

If we completely remove the content of Address 1: Street 1 and save, we will be thrown a business process error with the message we have chosen.

The screenshot shows the Microsoft Dynamics 365 Expense System interface. The contact record for Avery Howard is displayed. A 'Business Process Error' dialog box is shown, indicating that 'Mandatory fields are missing'. The 'Address 1: Street 1' field is empty. The 'Recent' tab shows a meeting from Jeremy Johnson and an email from CDS System. The 'Assistant' section shows no notifications or suggestions. The 'Company' section shows Trey Research. The 'Recent Opportunities' section shows a 50 Cafe A-100 Automatic opportunity for \$95,000.00, dated 1/15/2024.

### 3. Managing system jobs

Suppose that we have a process running and some technical issue happens. How do we check what happened? The event will be saved as a system job.

Click on the settings icon in Dynamics 365 and go to Advanced Settings.

The screenshot shows the Dynamics 365 interface. On the left is a navigation pane with 'Contacts' selected. The main area displays the contact record for 'Avery Howard', a CEO at 'Trey Research'. The 'CONTACT INFORMATION' tab is active, showing fields like First Name, Last Name, Job Title, Account Name, Email, Business Phone, Mobile Phone, Fax, Preferred Method of Contact, and Address. To the right of the contact information is a 'Timeline' section showing recent activities, including meetings and emails. On the far right, the 'Settings' menu is open, showing options like General, About, Privacy & Cookies, Software license terms, More Settings, Personalization Settings, Advanced Settings, Toast Notification Display Time, and Optional experiences.

The screenshot shows the Dynamics 365 'Settings' page for 'Business Management'. At the top, there is a warning banner: 'Web client experience for security settings is being deprecated'. Below this, the page is titled 'Business Management' and asks 'Which feature would you like to work with?'. There are two columns of feature tiles, each with an icon and a brief description:

- Fiscal Year Settings**: Set the start date, template, and display options for the fiscal year and fiscal period used for tracking sales goals.
- Business Closures**: Create a list of holidays and other times when the business is closed.
- Queues**: Create and manage service queues, and manage the membership of private queues. Establish criteria for automatic record creation and updates.
- Sales Territories**: Create new sales territories and assign territory managers. Add and remove members, modify territory information, and delete territories.
- Sites**: Create new sites or office locations where service operations take place. Add and remove resources, change site information, or delete sites.
- Currencies**: Add new currencies or change the exchange rates for existing currencies.
- Automatic Record Creation and Update Rules**: Create and manage rules for automatic record creation and updates. You can set up rules for either out-of-the-box entities or custom entities.
- LinkedIn Sales Navigator**: Manage settings relating to LinkedIn Sales Navigator integration.
- Goal Metrics**: Define and manage the kinds of goals that your organization tracks.
- Facilities/Equipment**: Add facilities and equipment for service scheduling. Change information about resources or delete existing resources.
- Resource Groups**: Add new groups and new members to existing groups for service scheduling. Update group information and delete groups or group members.
- Services**: Add new services for service scheduling. Change service information and deactivate existing services.
- Subjects**: Manage the subject hierarchy for your organization's products, literature, and articles.
- Connection Roles**: Create, edit, and delete the standard labels used to define connections between records.
- Rollup Queries**: Go to your list of Rollup Queries that you can use to gather data about a group of related records.

## Navigate to System Jobs.

The screenshot shows the Dynamics 365 Settings page. The top navigation bar includes 'Dynamics 365', 'Settings', and 'Business Management'. Below the navigation bar, there are several tabs: Business, Customization, System, Process Center, Application, and Upgrade Logs. The 'System' tab is selected, showing a list of system jobs including Administration, Security, Data Management, System Jobs, Document Management, and Auditing. Other tabs like Sites, Currencies, Automatic Record Creation and Update Rules, and LinkedIn Sales Navigator are also visible.

## Here we will see all the system jobs with their status.

The screenshot shows the Dynamics 365 System Jobs page. The top navigation bar includes 'Dynamics 365', 'Settings', and 'System Jobs'. Below the navigation bar, there is a yellow banner for 'Enable Server-Based SharePoint Integration'. The main content area shows a list of system jobs with columns: System Job Name, Regarding, Status Reason, Owner, and Started On. The list contains 2475 jobs, all of which are 'Succeeded'.

| System Job Name                                                 | Regarding                                      | Status Reason | Owner           | Started On           |
|-----------------------------------------------------------------|------------------------------------------------|---------------|-----------------|----------------------|
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:26 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:26 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:26 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:26 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | SYSTEM          | 12/27/2023 2:24 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | SYSTEM          | 12/27/2023 2:24 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:22 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:22 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:22 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:20 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | SYSTEM          | 12/27/2023 2:20 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | SYSTEM          | 12/27/2023 2:20 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:16 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:16 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:16 P... |
| Calculate Rollup...                                             | Calculate rollup fields for the account entity | Succeeded     | SYSTEM          | 12/27/2023 2:16 P... |
| System Event: Microsoft.Dynamics.MicrosoftFlow.Plugins.Async... | Update Street A                                | Succeeded     | Nicola Paracone | 12/27/2023 2:13 P... |

Alternatively, we can go to Power Apps, open the process and click on Process Sessions.

Power Apps

File Close Deactivate Show Dependencies Solution Layers Actions Help

Process: Update Street Address Working on solution: Demo

Process Sessions

Common

- Information
- Audit History
- Catalog Assignments
- Comments
- SLA Items
- PM Recordings
- Agent script steps
- Macro Run Histories
- timespents
- Entities
- Chatbot subcompone...
- Card

Process Sessions

Process Sessions Process Session Associated View

Search for records

| Name     | Process Name (Process) | Category (Process) | Owner            | Status Reason | Started On      | Started By       | Regarding   |
|----------|------------------------|--------------------|------------------|---------------|-----------------|------------------|-------------|
| Workflow | Update Street Address  | Workflow           | Nicola Paraco... | Failed        | 12/27/2023 2... | Nicola Paraco... | Avery Ho... |
| Workflow | Update Street Address  | Workflow           | Nicola Paraco... | Failed        | 12/27/2023 2... | Nicola Paraco... | Avery Ho... |
| Workflow | Update Street Address  | Workflow           | Nicola Paraco... | Failed        | 12/27/2023 2... | Nicola Paraco... | Avery Ho... |
| Workflow | Update Street Address  | Workflow           | Nicola Paraco... | Failed        | 12/27/2023 2... | Nicola Paraco... | Avery Ho... |
| Workflow | Update Street Address  | Workflow           | Nicola Paraco... | Failed        | 12/27/2023 2... | Nicola Paraco... | Avery Ho... |
| Workflow | Update Street Address  | Workflow           | Nicola Paraco... | Failed        | 12/27/2023 2... | Nicola Paraco... | Avery Ho... |
| Workflow | Update Street Address  | Workflow           | Nicola Paraco... | Failed        | 12/27/2023 2... | Nicola Paraco... | Avery Ho... |

1 - 7 of 7 (0 selected)

Page 1

Status: Activated

File Close Actions Help

Process Session: Workflow

Information

Process Session : Informat...

General

Name: Workflow Process: Update Street Address

Regarding: Avery Howard Created On: 12/27/2023 2:29 PM

Owner: Nicola Paracone Status Reason: Failed

Summary

Step description: None provided.

If ContactAddress 1: Street does not contain data OR ContactAddress 1: City does not contain data OR ContactAddress 1: State/Province does not contain data, then:

Stop workflow with status of: Canceled View properties

Update: Contact View properties

Comments

Sync workflow failed with error message: System.Activities.Statements.WorkflowTerminatedException: Mandatory fields are missing --> Microsoft.Xrm.Sdk.InvalidPluginExecutionException

End of inner exception stack trace -->

at System.Activities.WorkflowApplication.Invoke(Activity activity, IDictionary`2 inputs, WorkflowInstanceExtensionManager extensions, TimeSpan timeout)

at System.Activities.WorkflowInvoker.Invoke(Activity workflow, IDictionary`2 inputs, TimeSpan timeout, WorkflowInstanceExtensionManager extensions)

at Microsoft.Crm.Workflow.SynchronousRuntime.SynchronousWorkflowActivityHost.ExecuteWorkflowUsingInvoker(Activity workflow, CommonWorkflowContext context)

at Microsoft.Crm.Workflow.SynchronousRuntime.SynchronousWorkflowActivityHost.StartWorkflow(WorkflowActivationData activationData, ICommonWorkflowContext context, error code -- 2147220970)

Linked Sessions

Originating Session: Previous Linked Session:

Next Linked Session:

Details

Started On: 12/27/2023 2:29 PM Started By: Nicola Paracone

Completed On: 12/27/2023 2:29 PM Completed By:

Canceled On: Canceled By:

Status: Complete

Be careful that status records are deleted after 30 days (otherwise database storage would be filled soon).

Navigate to Data Management and to Bulk Record Deletion to delete system jobs.

Dynamics 365

Settings

Business Management

Settings

Business

Business Management...

Templates

Product Catalog

Service Management

Mobile Offline

Sync Error

Customization

Customizations

Solutions

Microsoft AppSource

Plug-In Trace Log

Solutions History

System

Administration

Security

Data Management

System Jobs

Document Manage...

Auditing

Process Center

Processes

Microsoft Flows

Application

Apps

Upgrade Logs

Upgrade Runs

Sites

Create new sites or office locations where service operations take place. Add and remove resources, change site information, or delete sites.

Currencies

Add new currencies or change the exchange rates for existing currencies.

Automatic Record Creation and Update Rules

Create and manage rules for automatic record creation and updates. You can set up rules for either out-of-the-box entities or custom entities.

LinkedIn Sales Navigator

Manage settings relating to LinkedIn Sales Navigator Integration

Subjects

Manage the subject hierarchy for your organization's products, literature, and articles.

Connection Roles

Create, edit, and delete the standard labels used to define connections between records.

Rollup Queries

Go to your list of Rollup Queries that you can use to gather data about a group of related records.

**Dynamics 365** Settings Data Management

Enable Server-Based SharePoint Integration Enable Now

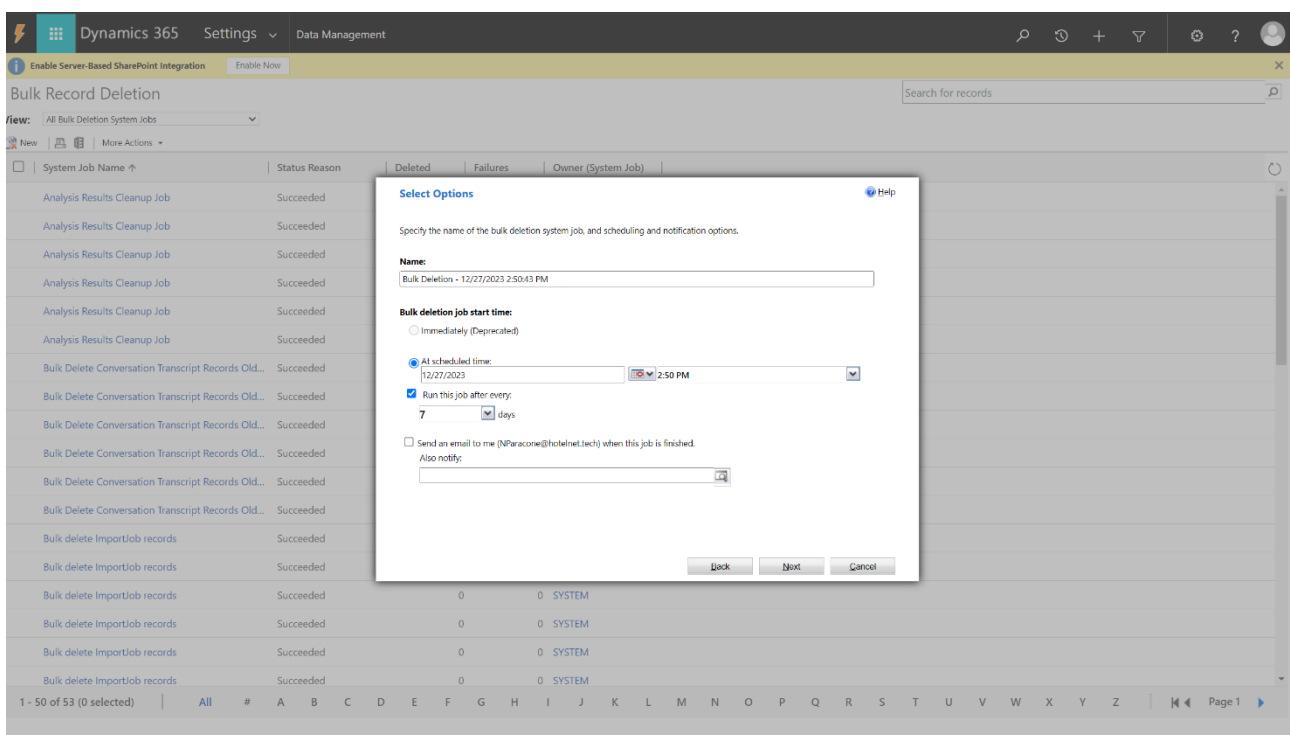
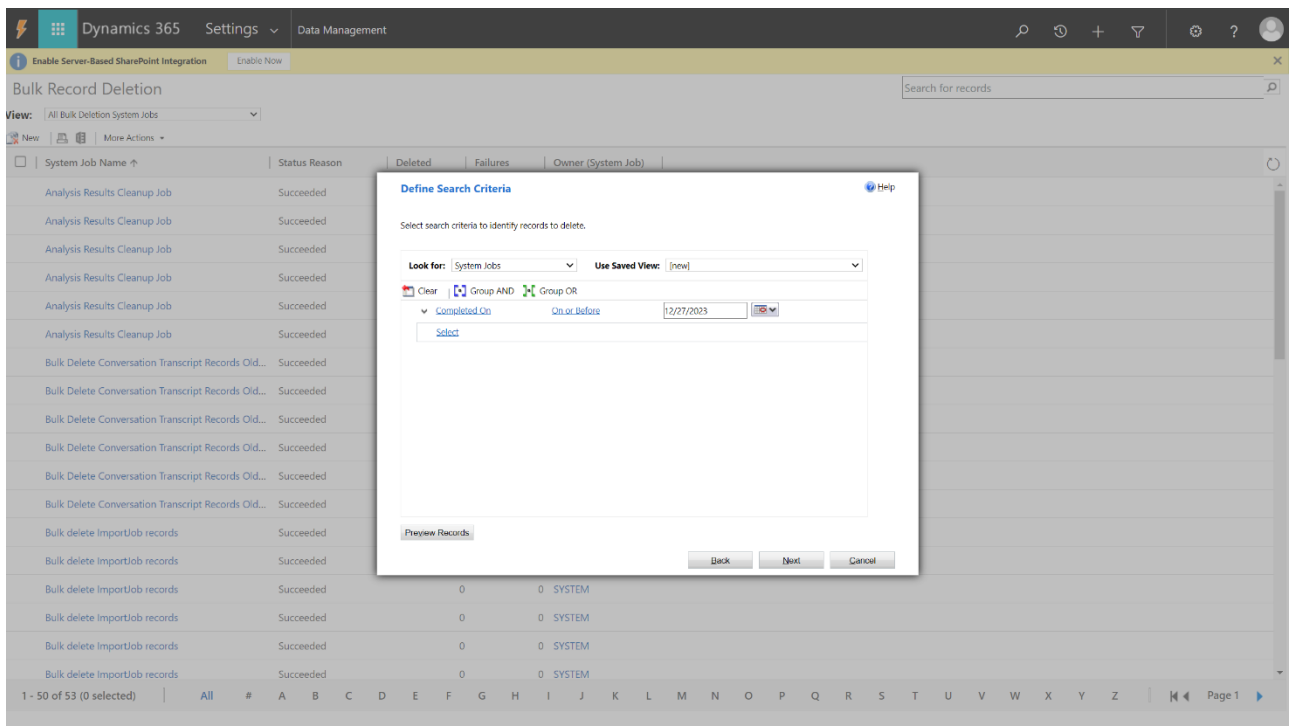
### Bulk Record Deletion

view: All Bulk Deletion System Jobs

New [Icons] More Actions

| System Job Name ↑                                  | Status Reason | Deleted | Failures | Owner (System Job) |
|----------------------------------------------------|---------------|---------|----------|--------------------|
| Analysis Results Cleanup Job                       | Succeeded     | 0       | 0        | SYSTEM             |
| Analysis Results Cleanup Job                       | Succeeded     | 0       | 0        | SYSTEM             |
| Analysis Results Cleanup Job                       | Succeeded     | 0       | 0        | SYSTEM             |
| Analysis Results Cleanup Job                       | Succeeded     | 0       | 0        | SYSTEM             |
| Analysis Results Cleanup Job                       | Succeeded     | 0       | 0        | SYSTEM             |
| Analysis Results Cleanup Job                       | Succeeded     | 0       | 0        | SYSTEM             |
| Analysis Results Cleanup Job                       | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk Delete Conversation Transcript Records Old... | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk Delete Conversation Transcript Records Old... | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk Delete Conversation Transcript Records Old... | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk Delete Conversation Transcript Records Old... | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk Delete Conversation Transcript Records Old... | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk delete ImportJob records                      | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk delete ImportJob records                      | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk delete ImportJob records                      | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk delete ImportJob records                      | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk delete ImportJob records                      | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk delete ImportJob records                      | Succeeded     | 0       | 0        | SYSTEM             |
| Bulk delete ImportJob records                      | Succeeded     | 0       | 0        | SYSTEM             |

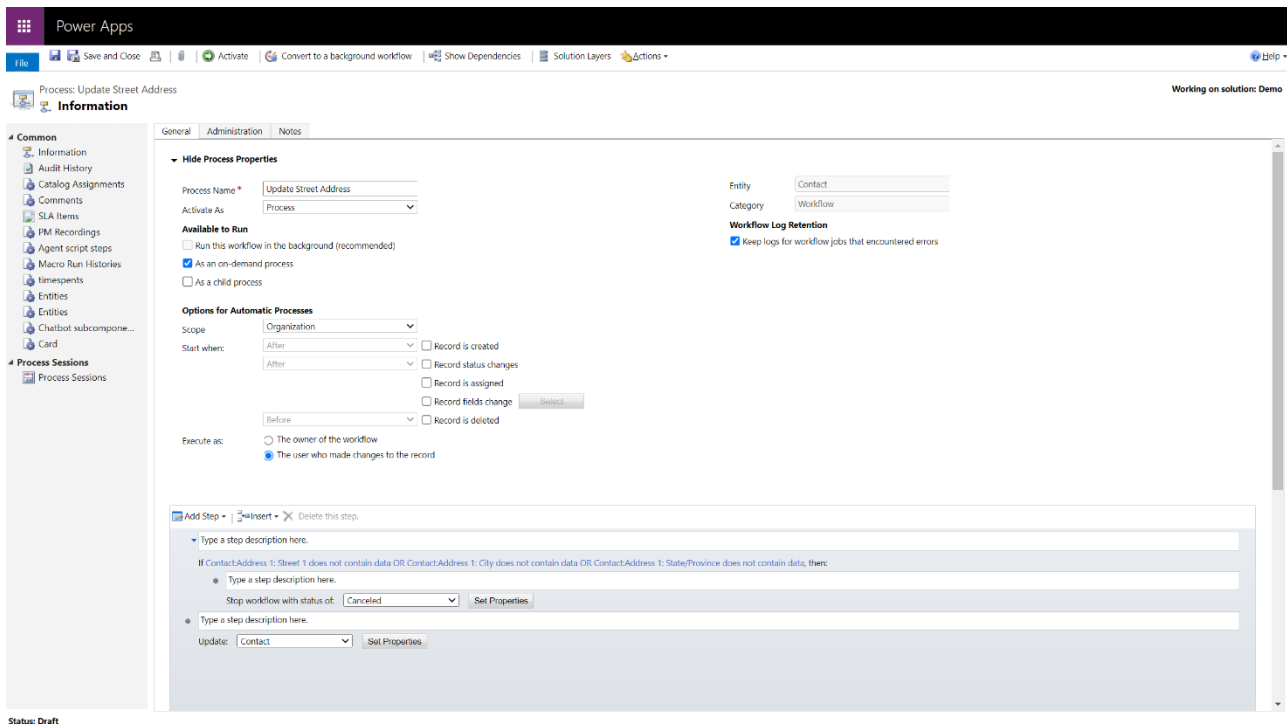
1 - 50 of 53 (0 selected) | All # A B C D E F G H I J K L M N O P Q R S T U V W X Y Z | Page 1



In the example above, every 7 days the job will run and delete the completed system jobs. This is to clean the system jobs to save storage.

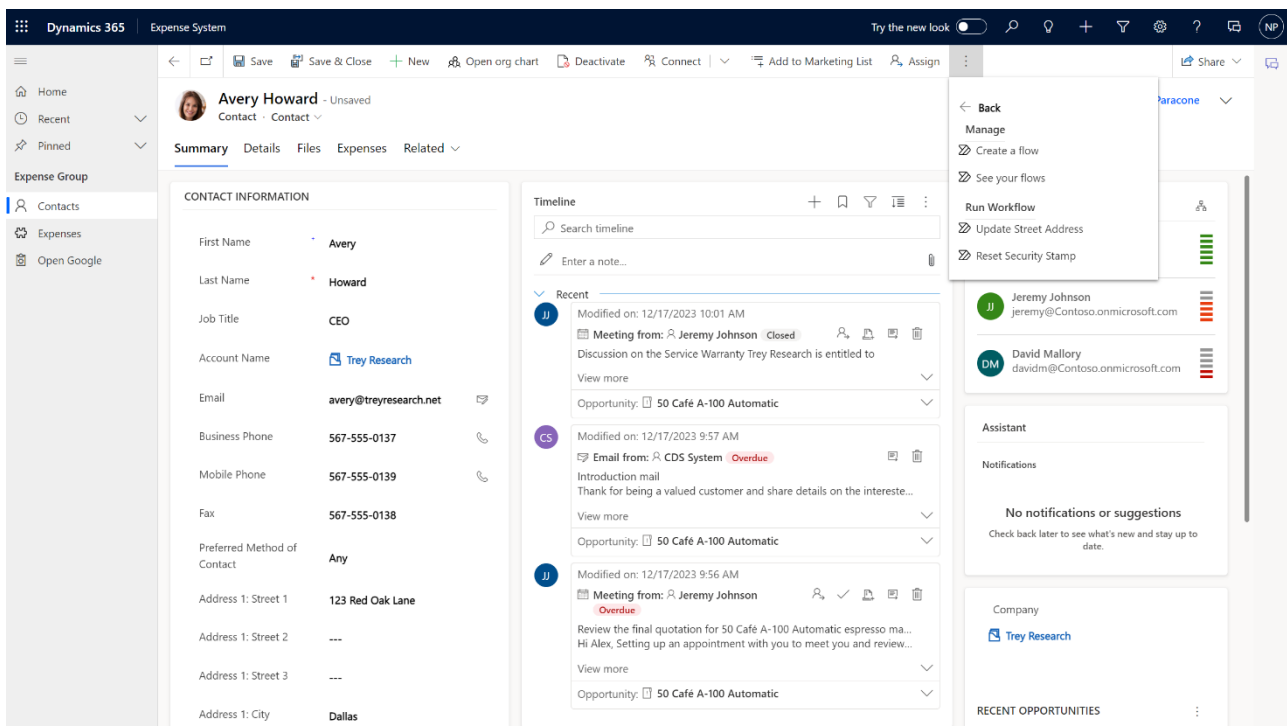


If we convert our real-time process to a background process, when we update the Address 1: Street 1 field we will see Address 1: Street 2 being populated only after a refresh. Remember that if the user needs to see the values immediately, then a real-time process is required. On more thing is that in the workflow designer we have the possibility to run the process as an on-demand process.



In this case the process has not dependencies on the fields that change (we uncheck the Record fields change box).

How are we going to access this process to run? Simply do as in the following figure.

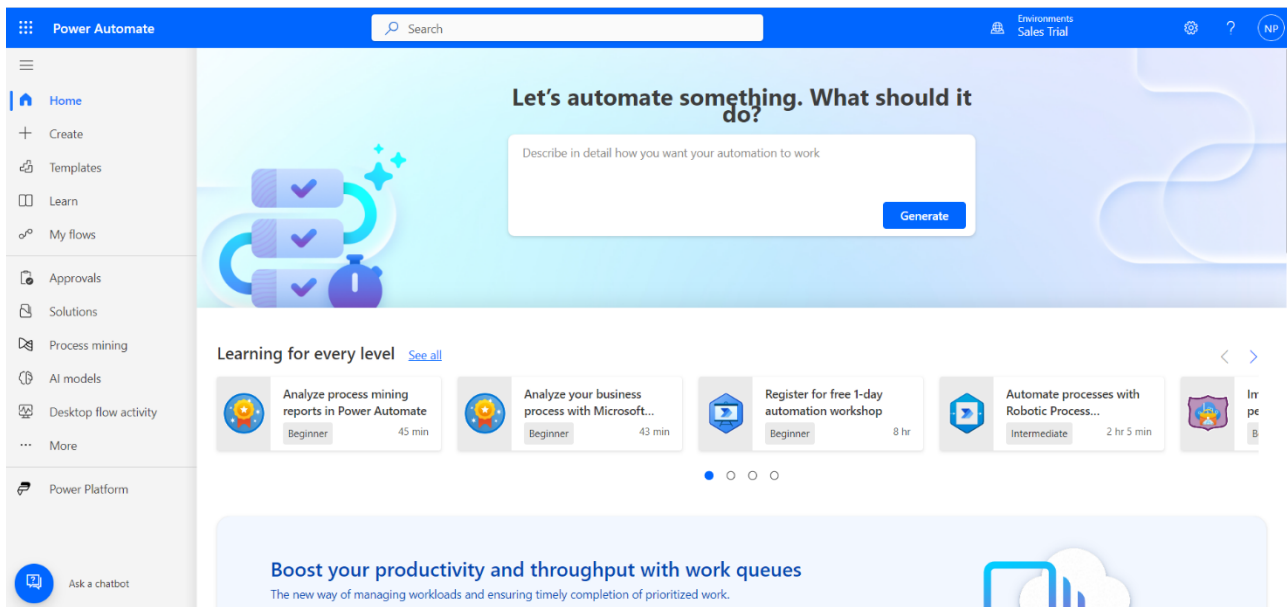


Clicking on our updater process, we will obtain the previous result.

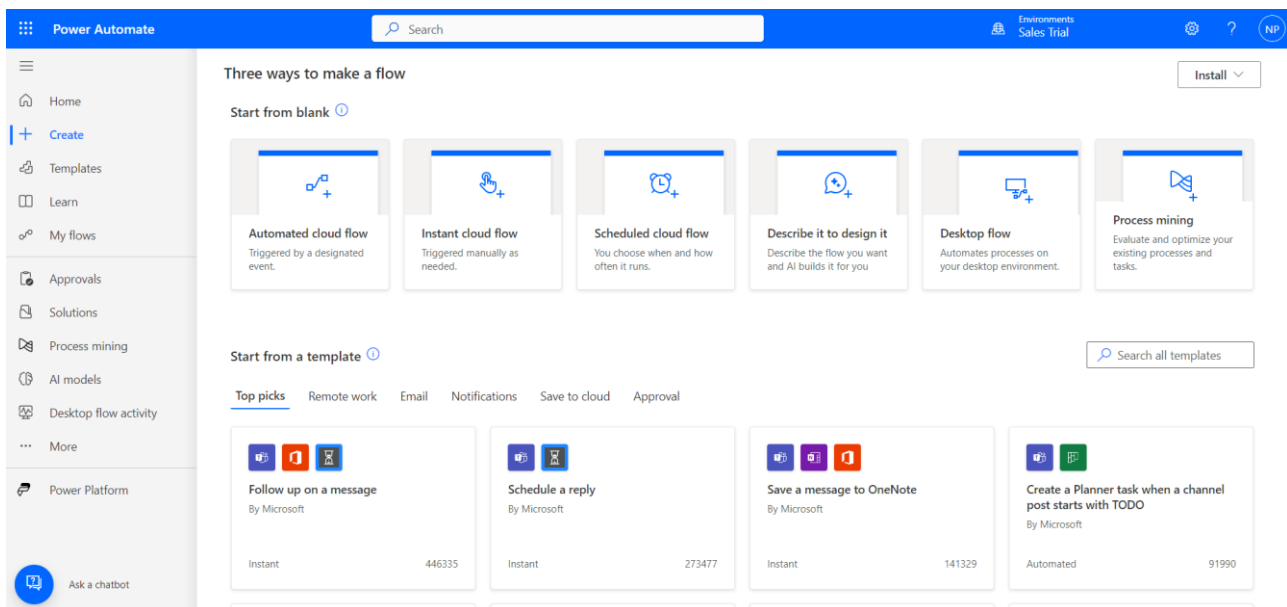
## 4. Create Power Automate flows

There are two ways of creating a Power Automate flow: either from Power Apps under Cloud Flows, or from Power Automate. In the second case we have full control.

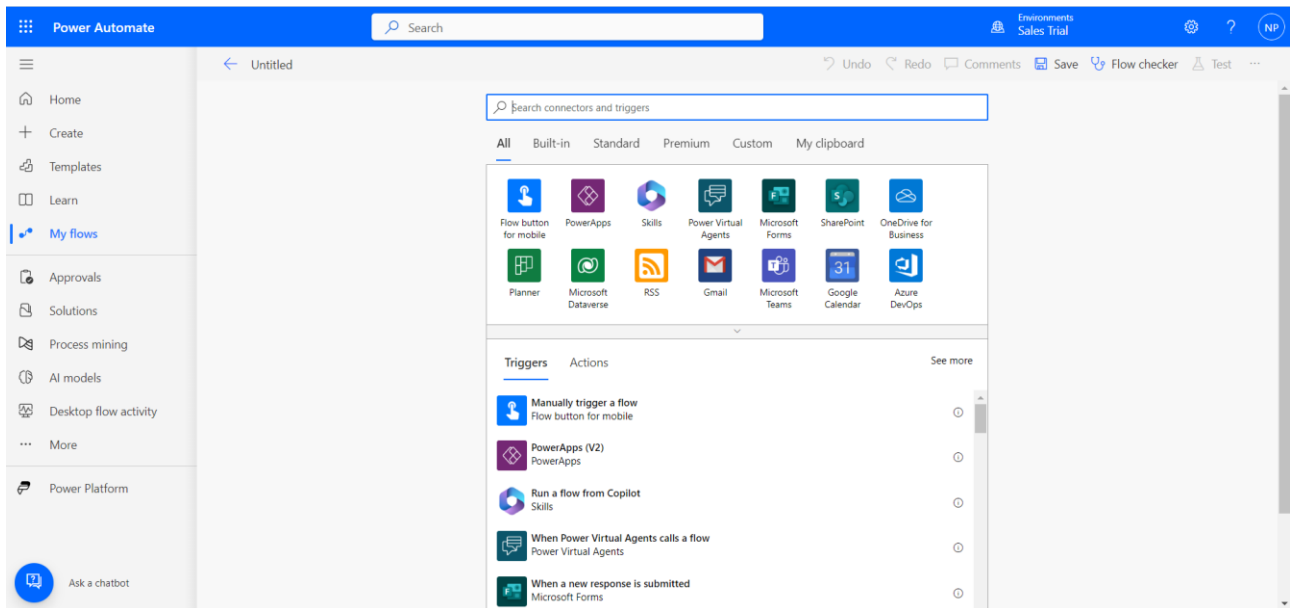
Navigate to [make.powerautomate.com](https://make.powerautomate.com). The interface is very similar to that of Power Apps.



To create a flow, we can do it from Solutions or by clicking + Create. The three flows we have are Automated cloud flow, Instant cloud flow and Scheduled cloud flow.

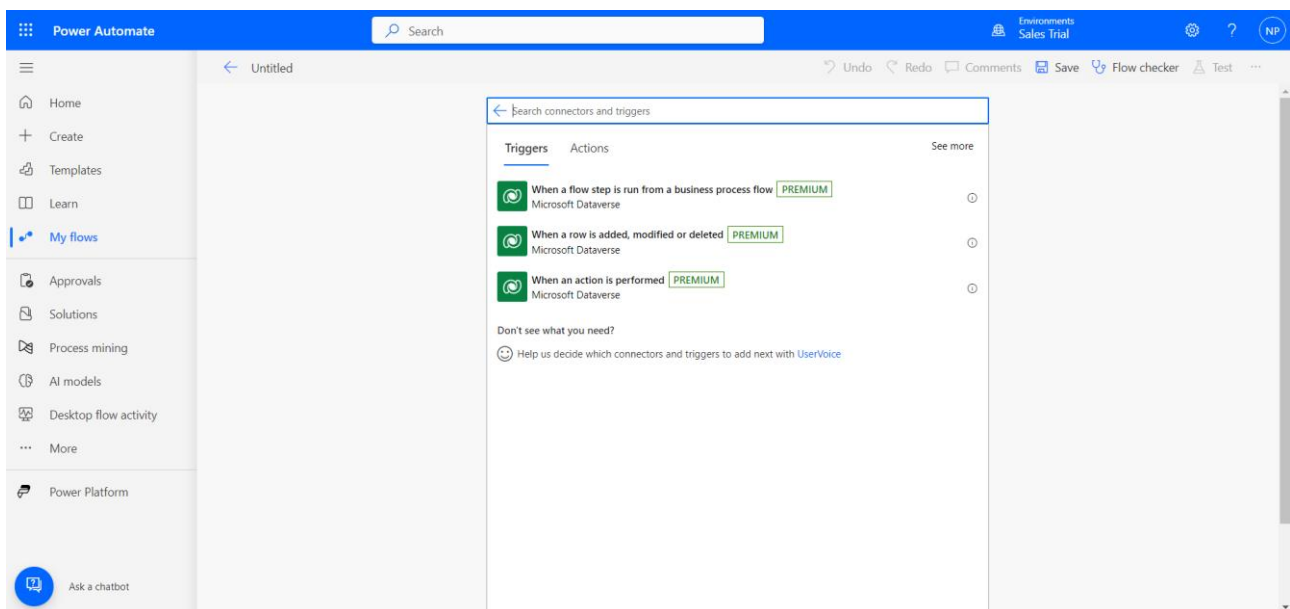


We start by creating an (untitled) Automated cloud flow. Click on Skip when the first creation window appears.

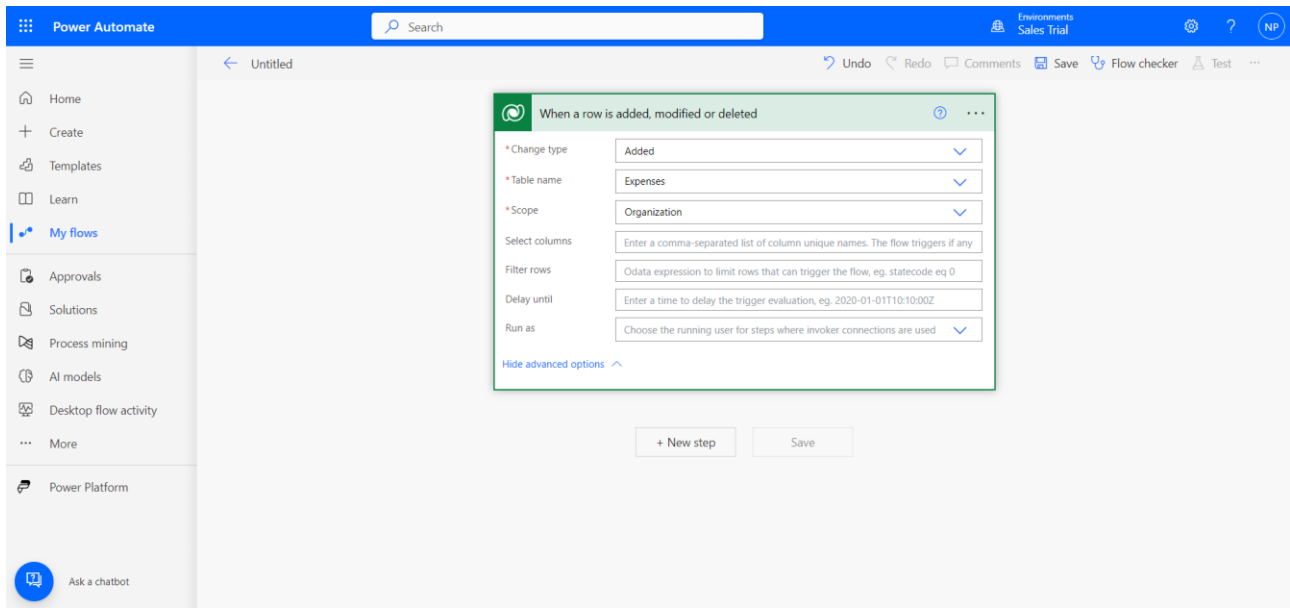


The Power Automate designer gives us many connectors, triggers and actions. Important connectors that should be explored are SharePoint, Gmail and Microsoft Teams. Most of the times, however, we will be using the Microsoft Dataverse connector.

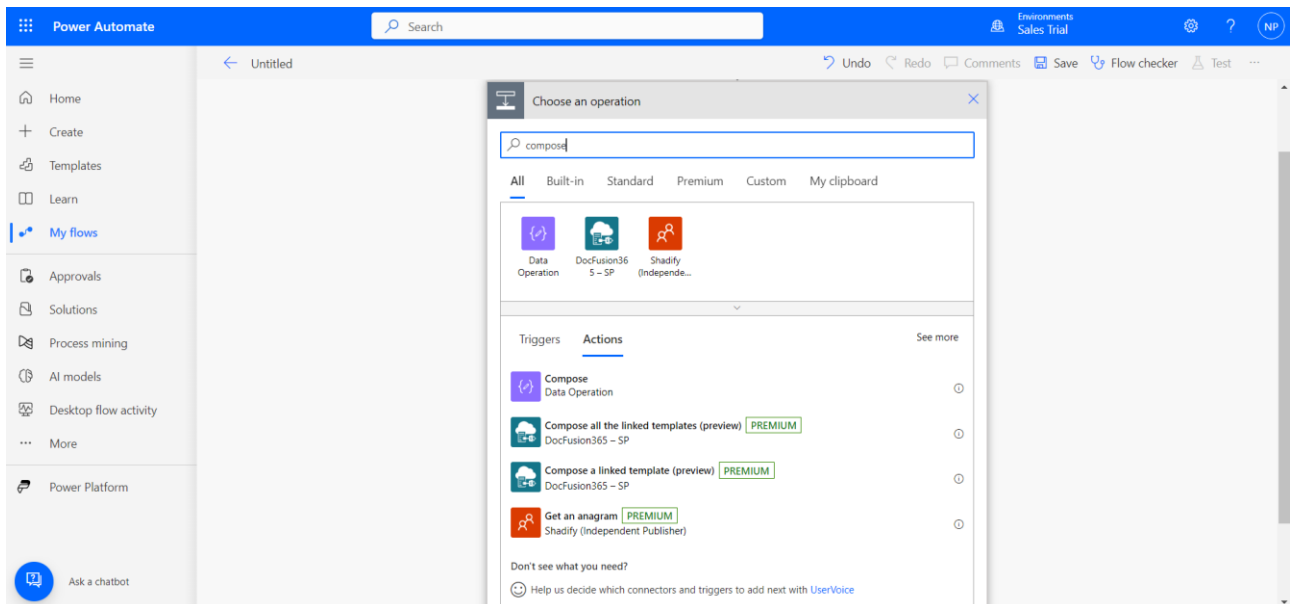
The trigger When a flow step is run from a business process flow means the following: we already saw a business process from lead qualification on. If the user is moving from one stage to the other, we can run our Power Automate flow in the middle (each stage will have its own Id).



As trigger we choose When a row is added, modified or deleted. Our aim, in fact, is the following: when a new expense is created, we want to get the contact associated with the customer and send an email with the expense content.

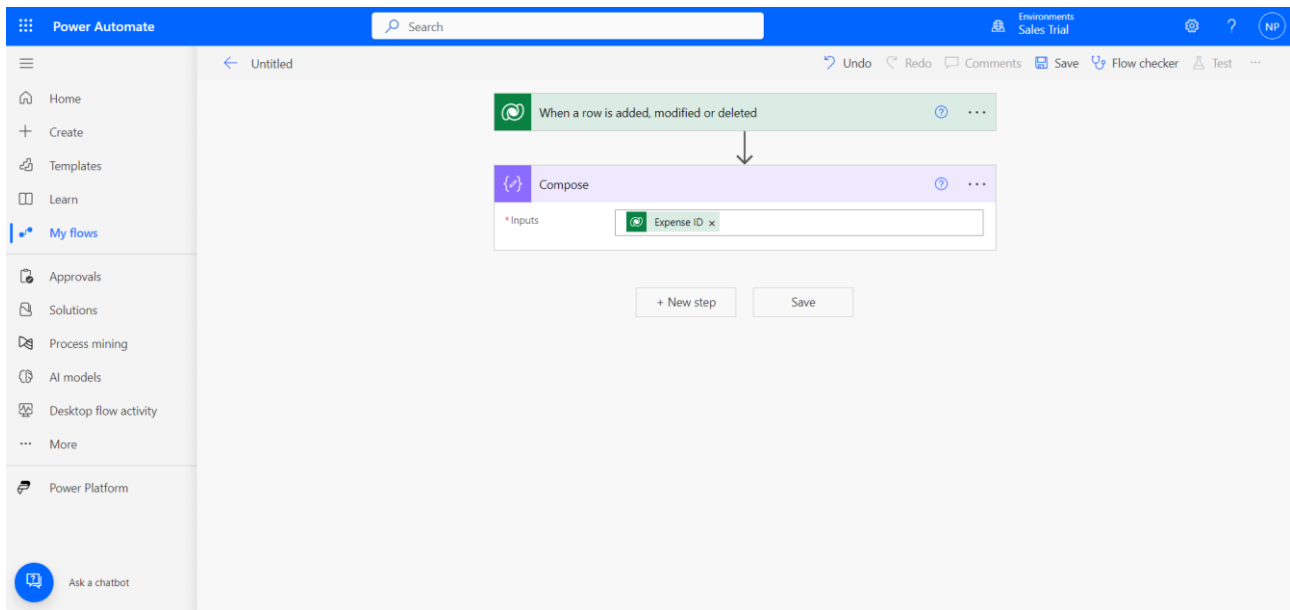


At the next step, search for "Compose" and under Actions select Compose.

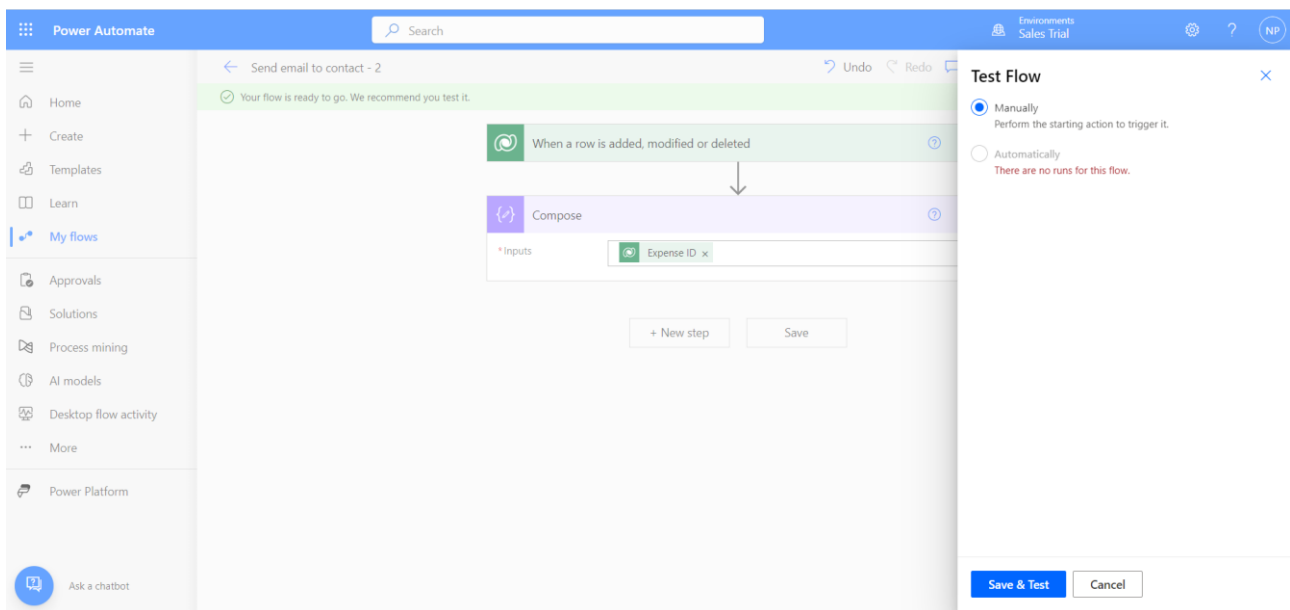


This action is commonly employed to manipulate and transform data before passing it to subsequent actions in the flow. It serves various purposes, such as data transformation, where complex expressions and formulas are used to modify received data. Additionally, Compose can be used to create variables within the flow, acting as temporary containers to store data for later use. It also plays a role in debugging by allowing users to inspect data during the flow-building process.

At this point, when a row (namely an expense) is added we need to get the expense Id.



Rename the flow as Send email to contact. Save the flow and click on Test (top right).



Now go to Expenses into Dynamics 365 and create (and save) a new expense for a contact.

The screenshot shows the Dynamics 365 Expense System interface. The left sidebar contains navigation options: Home, Recent, Pinned, Expense Group, Contacts, Expenses (selected), and Open Google. The main area displays the 'EXPENSE-100003' record, which has been saved. The record details are as follows:

| Field          | Value           |
|----------------|-----------------|
| Expense ID     | EXPENSE-100003  |
| Expense Type   | Flight          |
| Expense Amount | €300.00         |
| Contact        | Avery Howard    |
| Status Reason  | Active          |
| Owner          | Nicola Paracone |

Below the record details is a 'Description' field with a placeholder '---'.

Back to Power Automate we will see the results of running our flow.

The screenshot shows the Power Automate interface. The top bar indicates 'Your flow ran successfully.' The flow is titled 'Send email to contact'. The 'Scope' step is expanded, showing the following JSON body:

```
{
  "_owningbusinessunit_value": "9fbd2113-5b9c-ee11-be37-000d3ab11a",
  "_owningbusinessunit_value@Microsoft.Dynamics.CRM.lookuplogicalname": "Business Units",
  "_owningbusinessunit_type": "businessunits",
  "demo_expensesid": "f7ec1d96-c1a4-ee11-be37-000d3adecfce",
  "statecode": 0,
  "_statecode_label": "Active",
  "statuscode": 1,
  "_statuscode_label": "Active"
}
```

The 'Compose' step is also expanded, showing the following inputs and outputs:

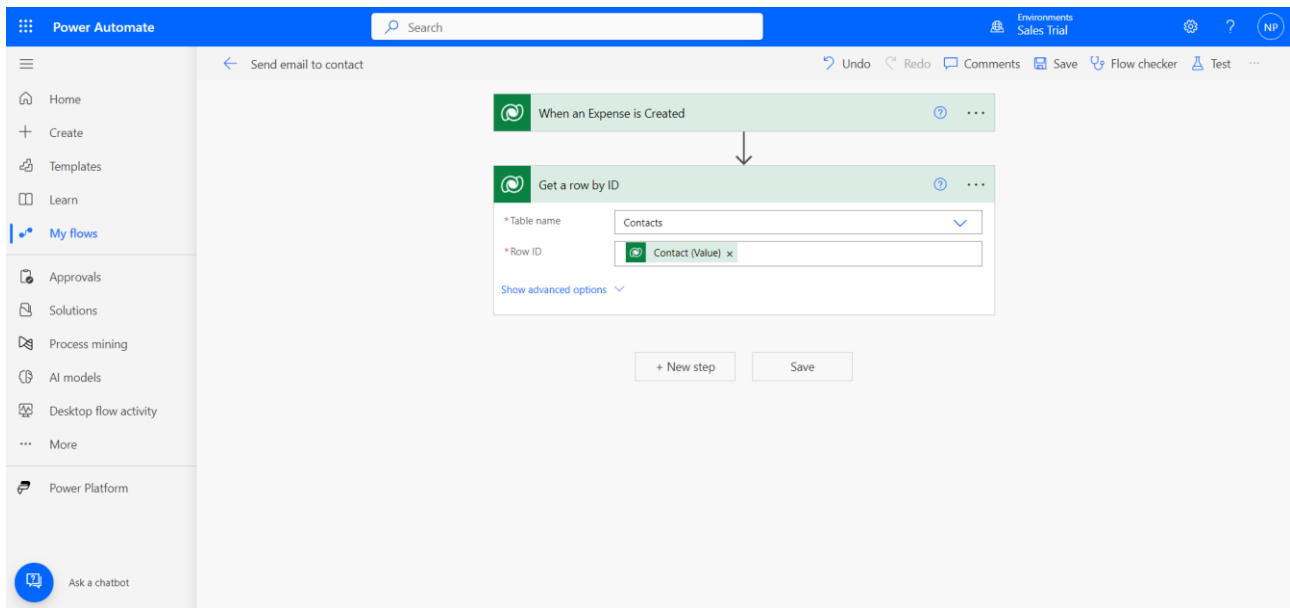
| Field   | Value          |
|---------|----------------|
| Inputs  | EXPENSE-100003 |
| Outputs | EXPENSE-100003 |

demo\_expensesid will be the same as the expense we created to test the flow.

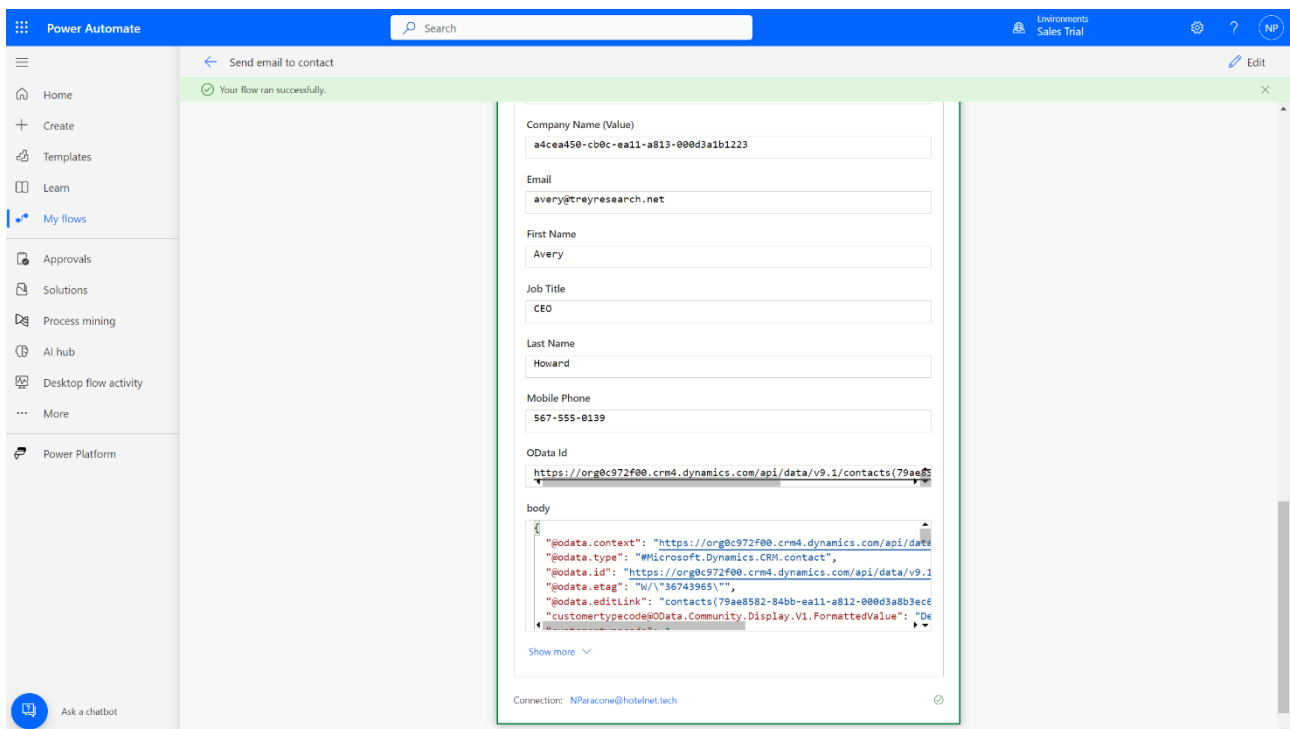
[https://org0c972f00.crm4.dynamics.com/main.aspx?appid=370541c5-01a4-ee11-be37-000d3adecfce&forceUCI=1&pagetype=entityrecord&etn=demo\\_expenses&id=f7ec1d96-c1a4-ee11-be...](https://org0c972f00.crm4.dynamics.com/main.aspx?appid=370541c5-01a4-ee11-be37-000d3adecfce&forceUCI=1&pagetype=entityrecord&etn=demo_expenses&id=f7ec1d96-c1a4-ee11-be...)

This was just a test. Continuing with the send email flow, what we need is to get the contact Id.

Modify the flow as in the following figure.

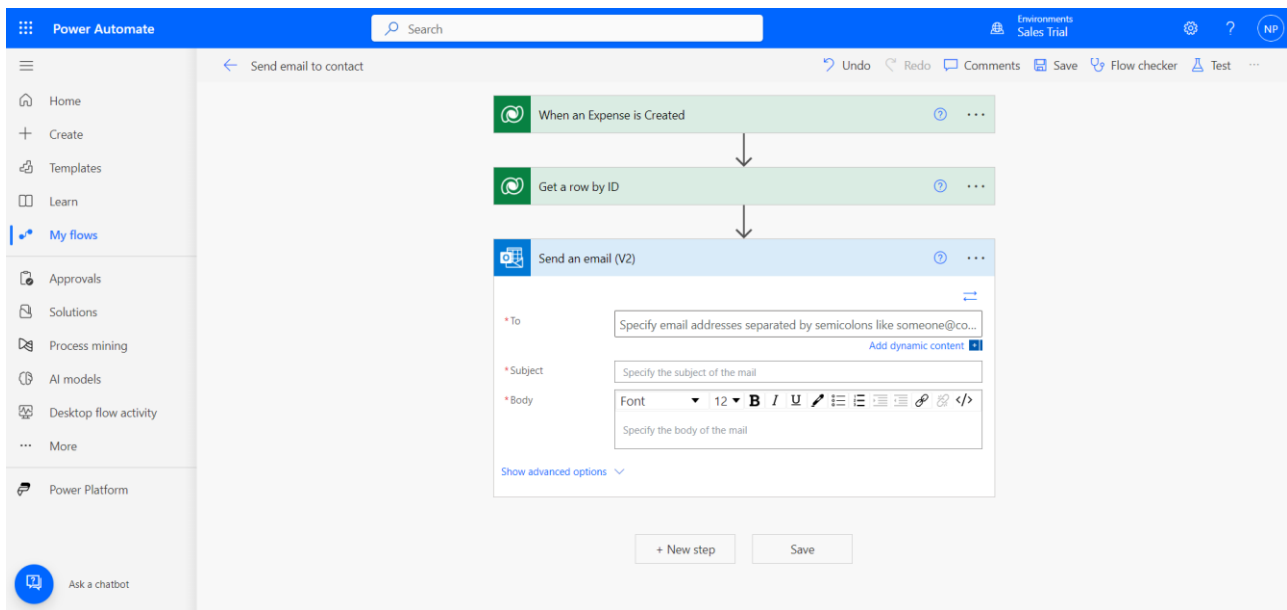


Contact (Value) will contain the GUID value of the contact. Save, test (automatically) and check the result.

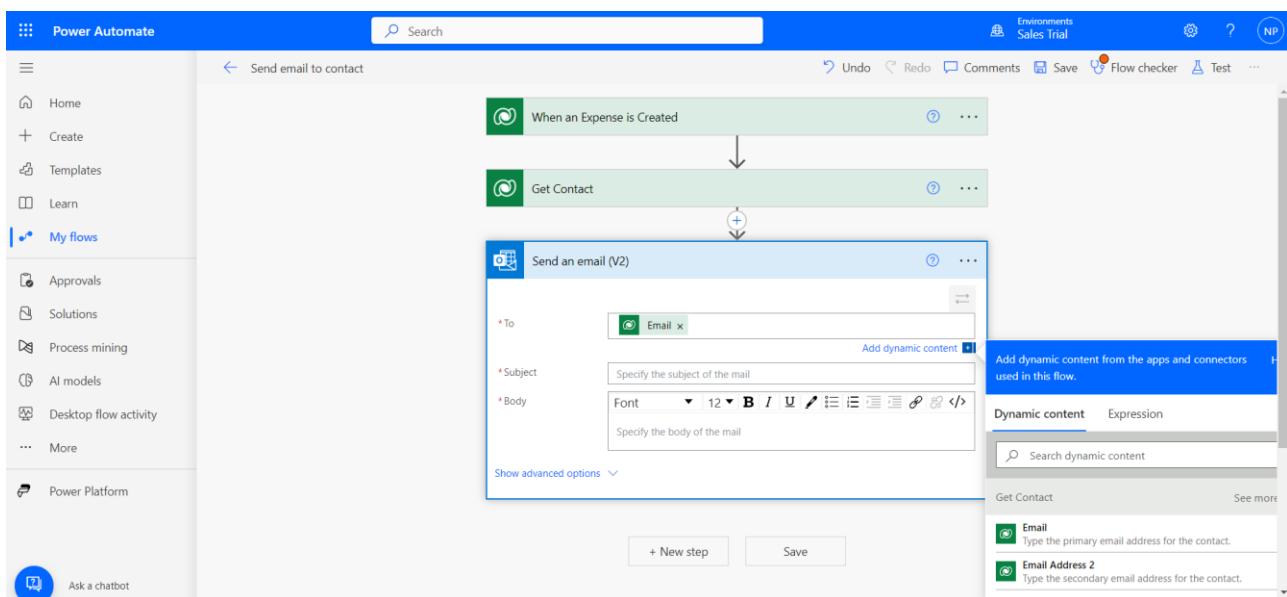


We add a new step to actually send an email.

Use Office 365 Outlook as connector and Send email (V2) as action.



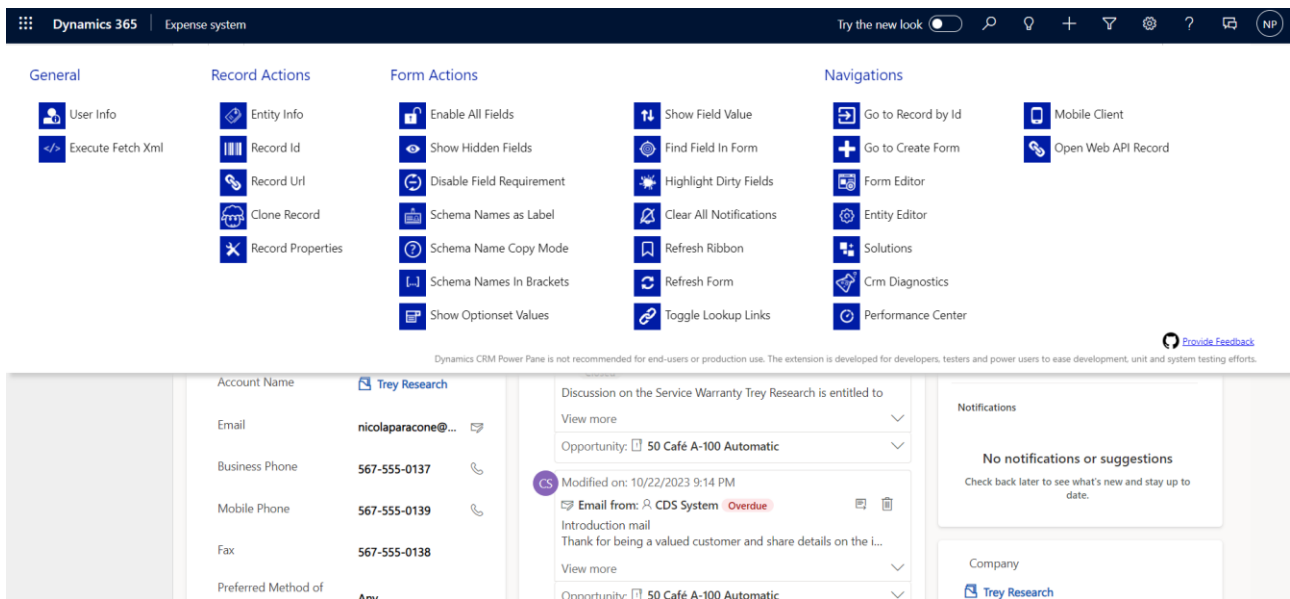
Click on Add dynamic content and choose Email.



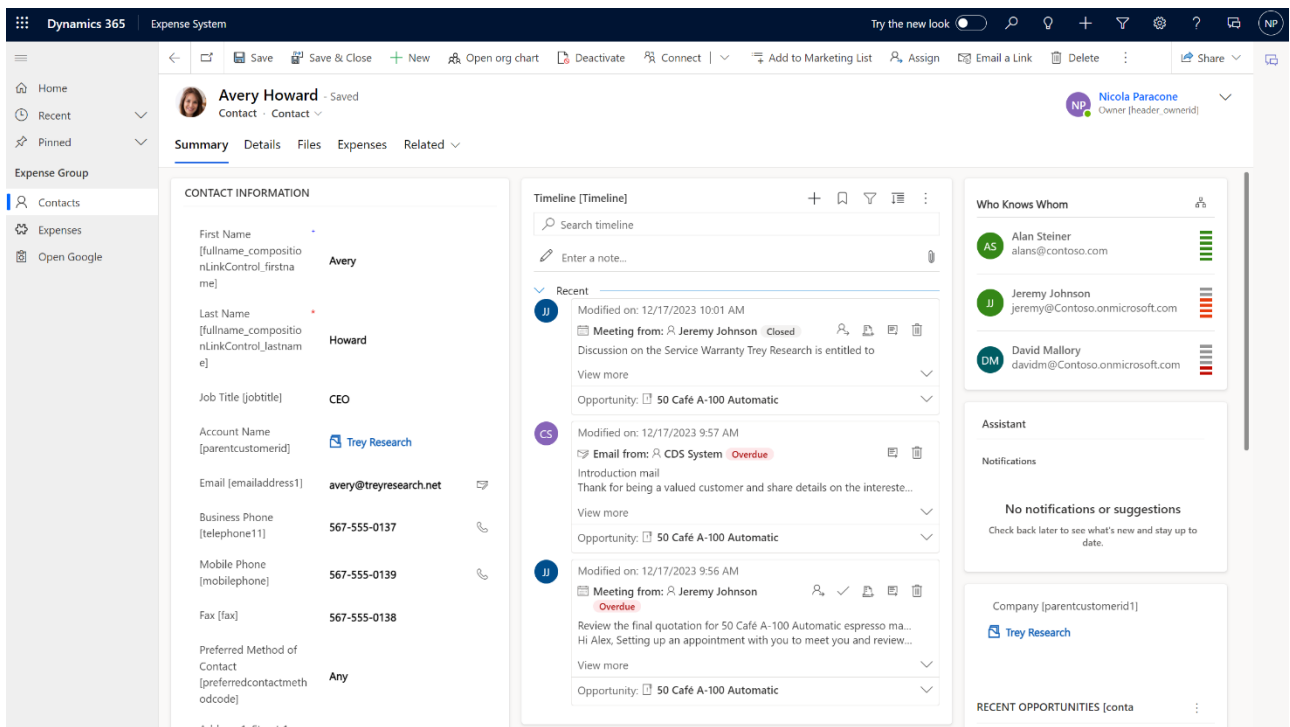
Then we do similarly to fill the body of the email. When we have a contact open in Dynamics 365, we see only the names of the fields. There is the possibility of viewing the logical names of such fields by using a Google extension called Dynamics 365 Power Pane; one more extension is Level Up for Dynamics 365/Power Apps.



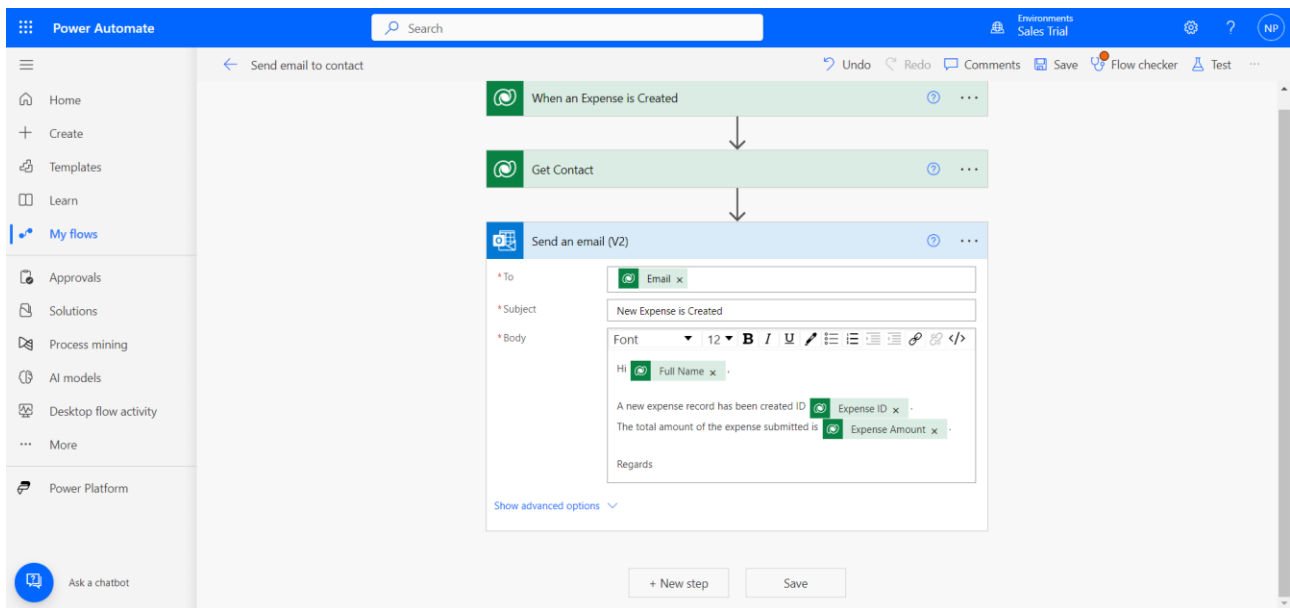
Once added, we click on the Power Pane icon and select Schema Names in Brackets.



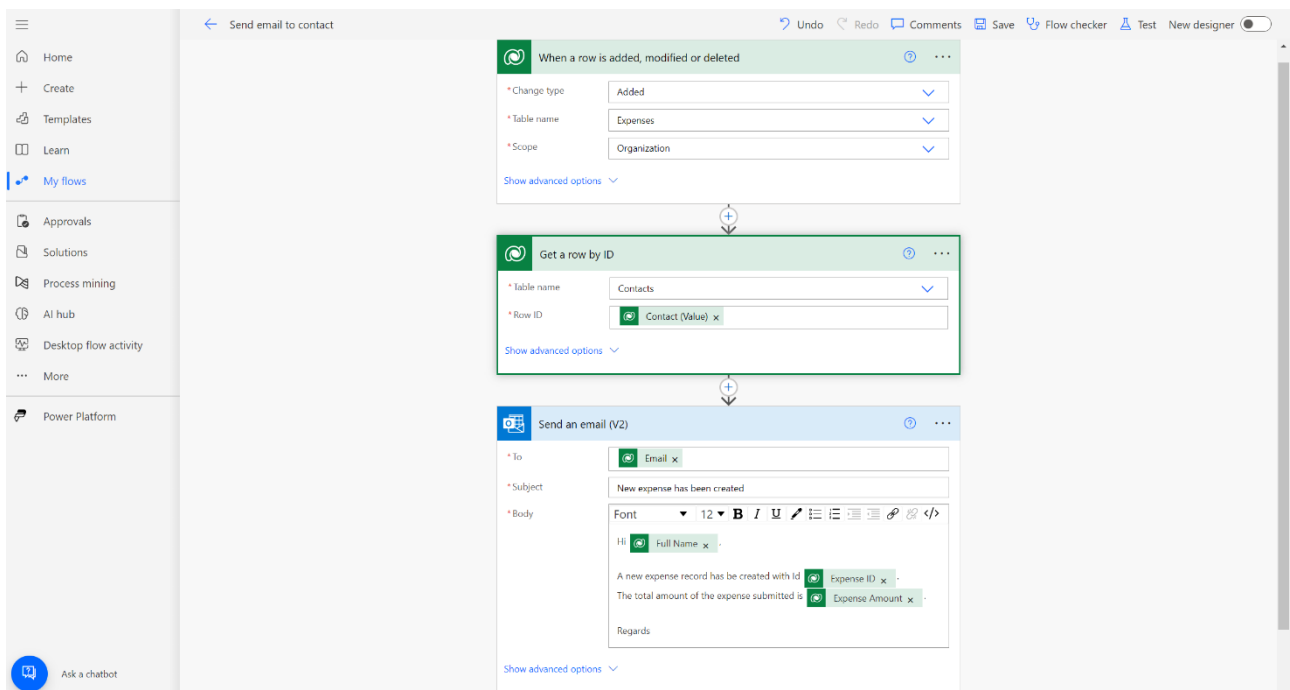
Then the logical name will appear next to the name of any field.



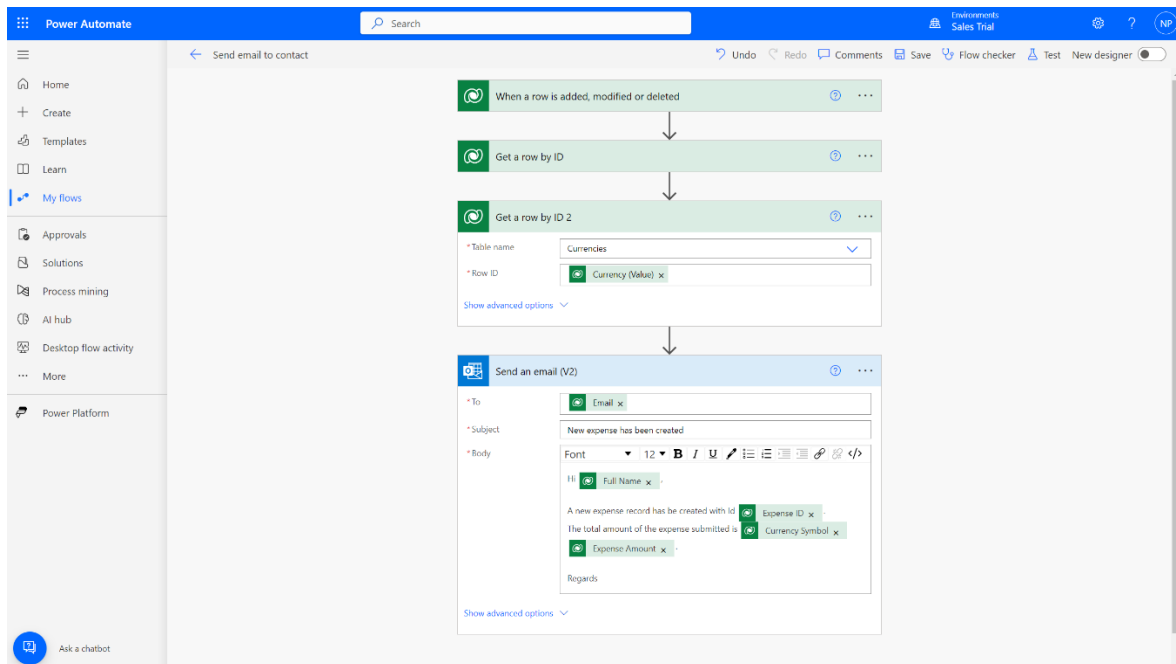
Remember that Microsoft does not allow to create two things with the same logical name.



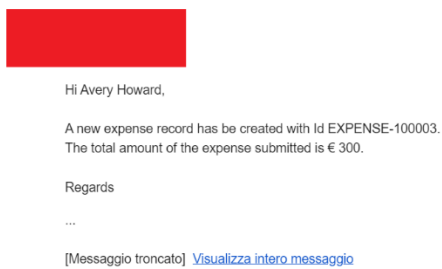
In some cases, we may like to add some style to the email (click on `</>`).



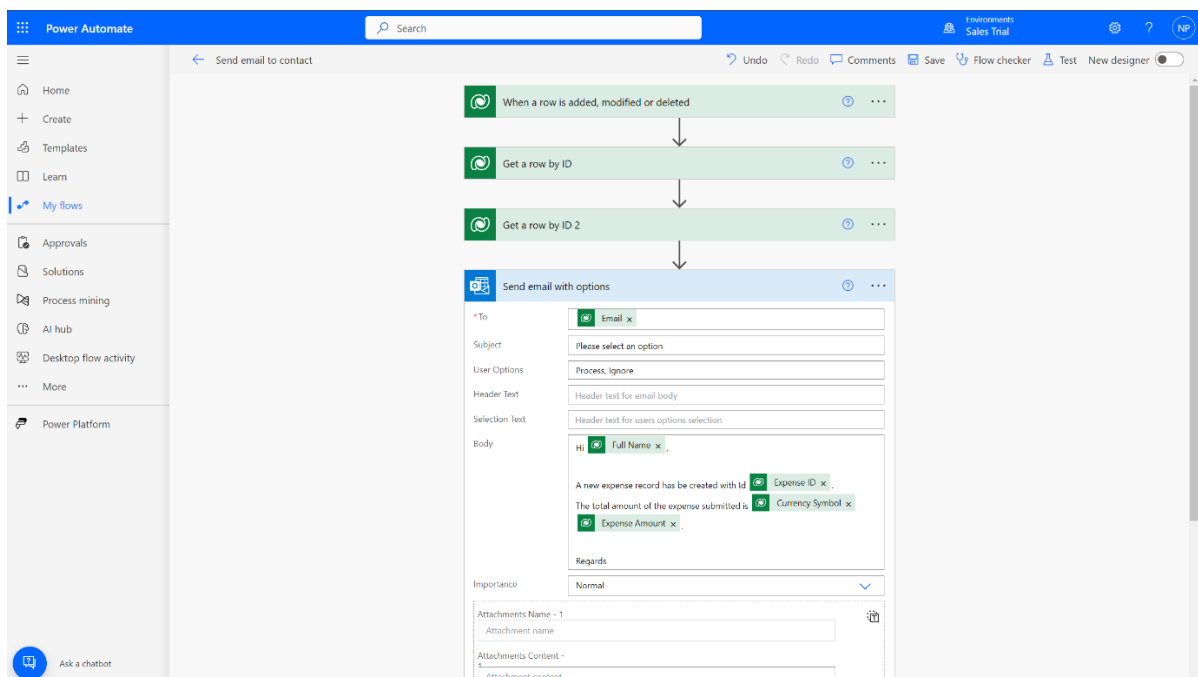
For clarity, we can also add the currency symbol.



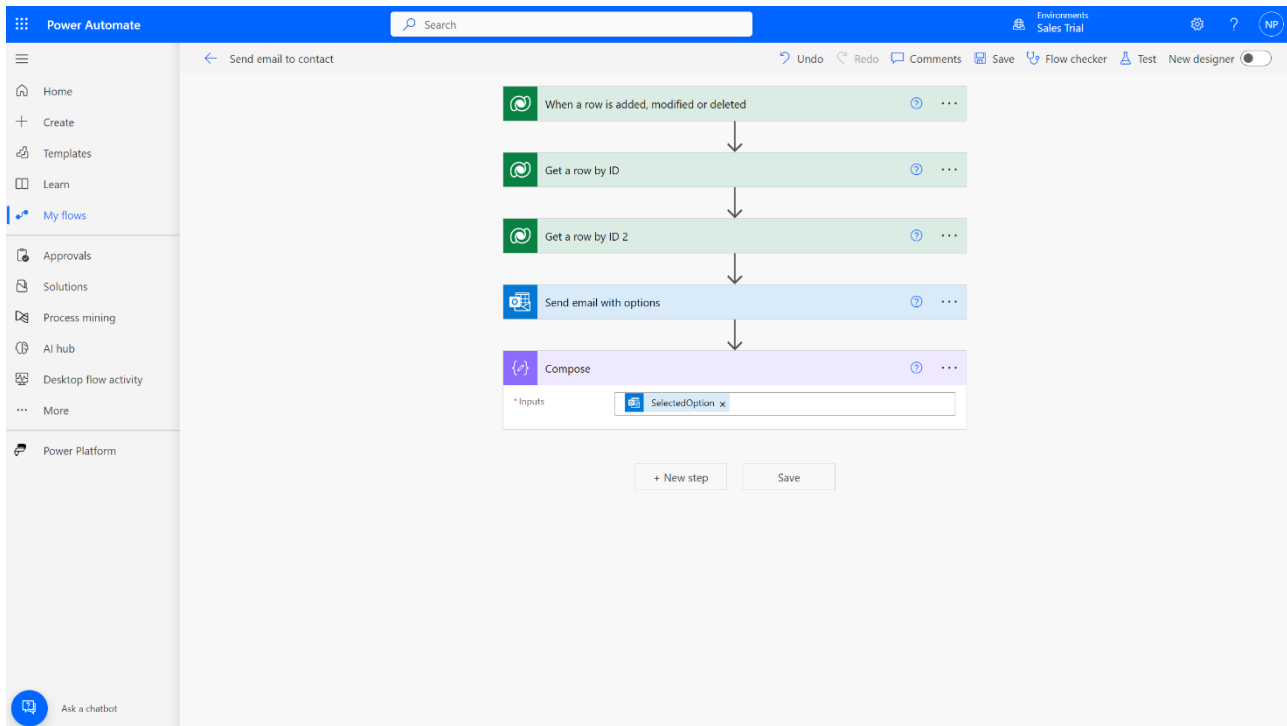
The result of testing this flow on a recently used trigger will be the following.



Assume now that the recipient must answer by choosing an option in the email. For that we use Send email with options.

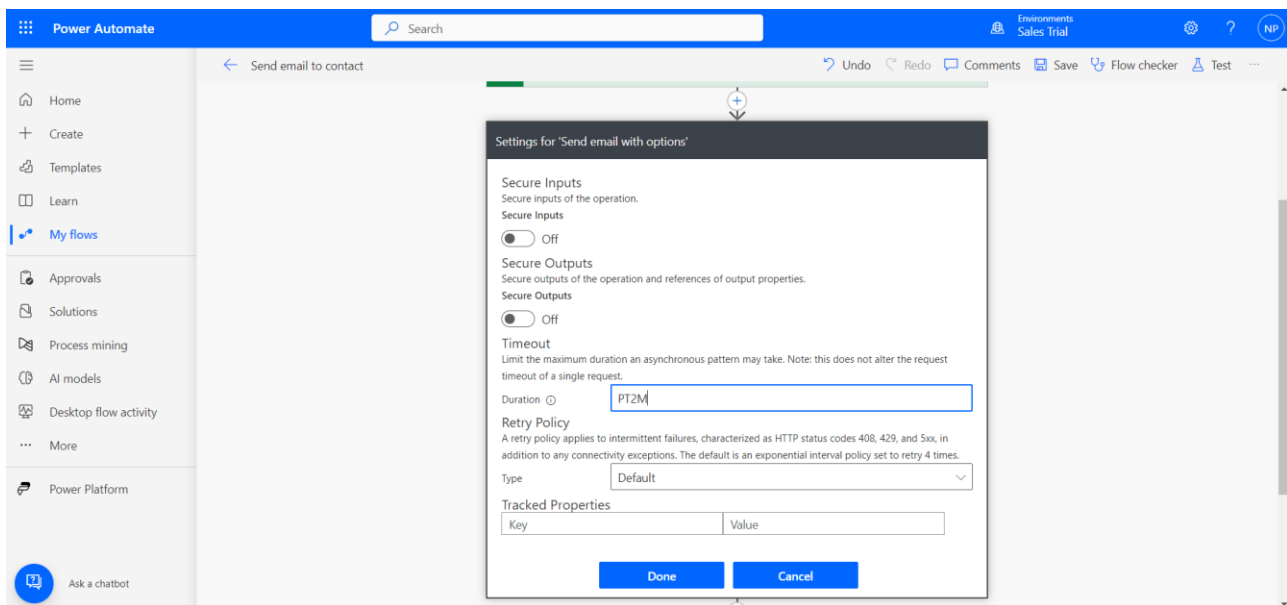


Using Compose (used when we want to store data without creating any new variables), we can store the option that the user has chosen.

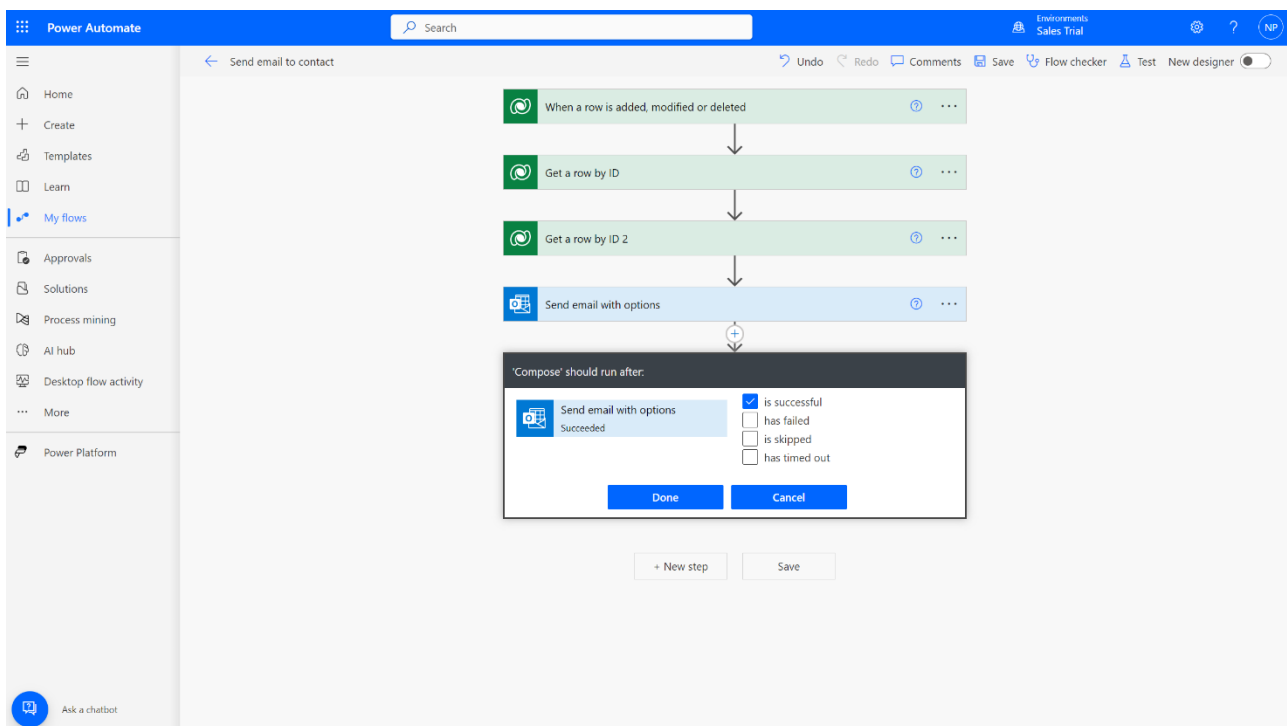
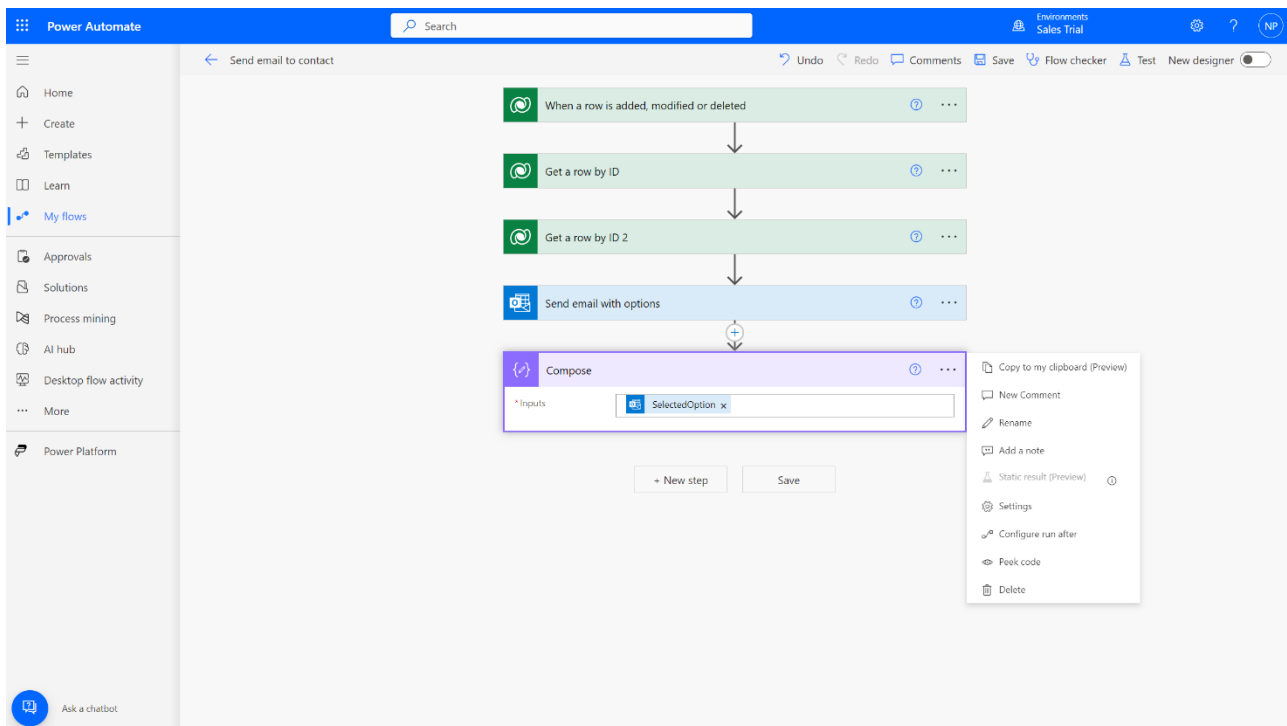


In general, the run will be active until the user selects one option. Obviously, we cannot accept that.

Hence, we access the settings of the step and set up a Duration time.



For example, P1D is one day, P2M is two months, and PT2M is two minutes. Next, we configure the Compose to run only if the previous block has run successfully.



This will be the email the contact will receive.

### Request for your input

Hi Avery Howard, A new expense record has be created with Id EXPENSE-100004. The total amount of the expense submitted is € 2000. Regards

Select one of the options below to respond

[Process](#) [Ignore](#)

Message sent via Microsoft Power Automate, enabling you to create automated workflows between your favorite apps and services.  
© Microsoft Corporation 2023

Additionally, we can insert a parallel branch. The left block will be executed if the Send email with options step is successful, the right block if it is unsuccessful.

